

MATHEMATICS GRADE 10: QUADRATIC EQUATIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Total Duration	7 lessons × 40 minutes = 280 minutes total
Sub-Strand Content	– Quadratic expressions: identifying, expanding, and simplifying expressions of the form $ax^2 + bx + c$ – Factorisation of quadratic expressions (including difference of two squares and perfect squares) – Forming quadratic equations from real-life contexts (e.g., area problems, Harambee fundraising targets) – Solving quadratic equations by factorisation – Solving quadratic equations by completing the square – Solving quadratic equations using the quadratic formula – Applications of quadratic equations in day-to-day situations
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify and write quadratic expressions of the form $ax^2 + bx + c$ in different situations, - expand and simplify quadratic expressions using algebraic rules, - factorise quadratic expressions including perfect squares and difference of two squares, - form quadratic equations from real-life contexts such as area, fundraising, and motion, - solve quadratic equations by factorisation, - solve quadratic equations by completing the square, - solve quadratic equations using the quadratic formula, - apply quadratic equations to solve problems in day-to-day situations, - appreciate the relevance of quadratic equations in modelling real-world phenomena.
Core Competencies	– Communication and Collaboration: The learner discusses with peers to expand, simplify, and factorise quadratic expressions, and shares solution strategies for equations derived from real-life Harambee fundraising and farming contexts. – Critical Thinking and Problem Solving: The learner analyses real-life situations (e.g., calculating the dimensions of a shamba plot in Kericho given its area) to form and solve quadratic equations, selecting the most efficient method among factorisation, completing the square, or the quadratic formula. – Digital Literacy: The learner uses calculators and digital devices to verify solutions to quadratic equations and explore graphical representations of quadratic functions. – Learning to Learn: The learner reflects on errors in algebraic manipulation and refines personal strategies for solving quadratic equations across multiple methods. – Self-Efficacy: The learner builds confidence by successfully modelling and solving progressively complex quadratic problems

	rooted in Kenyan everyday contexts.
Core Values	– Unity: The learner works collaboratively in groups to explore and compare different methods of solving quadratic equations, appreciating diverse approaches. – Responsibility: The learner takes responsibility for accuracy in algebraic working, understanding that errors in quadratic modelling (e.g., miscalculating a land area or fundraising shortfall) have real consequences. – Integrity: The learner honestly records and presents their working, showing all steps in solving quadratic equations rather than copying answers. – Social Justice: The learner recognises how mathematical literacy, including understanding quadratic relationships, empowers community members to make fair and informed decisions about resources and fundraising.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students pose mathematical questions about what dimensions or quantities satisfy a given quadratic constraint (e.g., "What side length gives exactly 96 m² for the Harambee hall floor?"). – Developing and Using Models: Students build and revise algebraic models (quadratic equations) to represent real-life situations, updating their models as new constraints are introduced. – Using Mathematics and Computational Thinking: Students apply factorisation, completing the square, and the quadratic formula as computational tools to obtain exact solutions, and use calculators to verify numerical answers. – Constructing Explanations: Students explain why a quadratic equation may have two, one, or no real solutions, connecting algebraic reasoning to the physical sense of the context (e.g., rejecting negative lengths). – Engaging in Argument from Evidence: Students compare solution methods and justify which method is most efficient for a given quadratic equation, supporting their argument with worked examples.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Critical Thinking and Problem Solving): The learner applies quadratic equations to analyse and solve real-life problems such as determining optimal dimensions for a community hall or maximising a farming plot's area in the Rift Valley. – Financial Literacy: The learner uses quadratic equations to model fundraising scenarios, such as calculating contribution amounts or shortfalls in a Harambee event, developing awareness of financial planning. – Environmental Education: The learner connects quadratic area models to land use and conservation contexts, such as calculating the area of irrigation plots along Lake Victoria to avoid over-use of resources. – Citizenship: The learner appreciates how mathematics supports equitable community development decisions, such as fair allocation of land or Harambee funds.
Career Connections	– Architecture and Civil Engineering: Quadratic equations are used to calculate structural dimensions, load distributions, and areas of buildings — directly relevant to constructing community halls, schools, and housing in Nairobi and beyond. – Agriculture and Land Management: Farmers and land surveyors in the Rift Valley use quadratic area models to optimise field layouts, irrigation coverage, and crop yield planning. – Finance and Accounting: Financial analysts and accountants use quadratic and polynomial models to project revenue targets, model loan repayments, and analyse cost-profit relationships. – Physics and Engineering: Quadratic equations model projectile motion, enabling careers in mechanical engineering, aerospace, and sports science (e.g., analysing the trajectory of a javelin thrown by a Kenyan athlete). – Data Science and Information Technology: Programmers and data scientists use quadratic and higher-order functions in algorithm optimisation, machine learning cost functions, and computer graphics rendering.
Focus for Lessons	This unit develops learners' ability to work fluently with quadratic expressions and equations, moving from conceptual understanding of the algebraic structure $ax^2 + bx + c$ through to practical solution methods and real-world application. Using the anchoring context of a community Harambee fundraising challenge in a Kenyan village, learners discover that some real-life problems cannot be solved with linear equations alone, motivating the need for quadratic thinking. Across seven lessons, learners progress from recognising and forming quadratic expressions, through three distinct solution methods (factorisation, completing the

	square, and the quadratic formula), to critically applying their skills in authentic Kenyan contexts such as land area calculations, construction planning, and financial modelling.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai? KICD KEY INQUIRY QUESTIONS: 1. How do quadratic expressions and equations differ from linear ones, and why are they necessary? 2. How can we use quadratic equations to model and solve problems in day-to-day activities?
Anchoring Phenomenon	A community in Githurai, Nairobi, is planning a Harambee to raise funds to build a new multipurpose hall. The organising committee knows the hall must have a rectangular floor with an area of exactly 120 m^2, and that the length must be 2 metres more than the width. When the committee tries to find the dimensions using simple arithmetic or a linear equation, they get stuck — no straightforward linear method gives them an answer. A local Form 4 student suggests the problem requires a quadratic equation. Students are shown the committee's hand-drawn sketch on a manila paper: a rectangle labelled "width = w" and "length = $w + 2$", with "Area = 120 m^2" written in the centre. The puzzle is immediately visible: how can you find a single unknown when it appears multiplied by itself? Students are asked to predict the width before any instruction is given, and most guesses are either wrong or arrived at by trial-and-error — revealing the limitation of their current tools and creating genuine need for the new mathematics of this unit.
Storyline Thread	Lesson 1 — PREDICT & ANCHOR PHENOMENON: Students examine the Githurai Harambee hall problem. They predict the hall dimensions using trial-and-error and discuss why linear equations fall short. The class co-constructs the equation $w(w + 2) = 120$, identifying it as a new type of equation. Students are introduced to the general form $ax^2 + bx + c = 0$ and begin recognising quadratic expressions in varied contexts. The Driving Question Board (DQB) is opened and initial "What we notice / What we wonder" cards are posted. Lesson 2 — EXPANDING & SIMPLIFYING QUADRATIC EXPRESSIONS: Students learn to expand products such as $(w + 2)(w + 3)$ and $(2x - 1)(3x + 4)$ using the distributive law and FOIL method. Supporting phenomena (the Kericho farm fencing problem) is introduced to show how real contexts generate quadratic expressions. Students simplify expressions and begin to see the structure of $ax^2 + bx + c$ emerging consistently. DQB is updated: "We now know how quadratic expressions are formed." Lesson 3 — FACTORISATION OF QUADRATIC EXPRESSIONS: Students reverse the expansion process to factorise quadratic expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$, including perfect square trinomials and difference of two squares. Using the Harambee hall equation ($w^2 + 2w - 120 = 0$), students practise factorising and identify factors that correspond to real dimensions. The Gikomba mandazi revenue problem is introduced as a factorisation challenge. DQB update: "Factorisation lets us 'undo' expansion to find hidden values." Lesson 4 — SOLVING QUADRATIC EQUATIONS BY FACTORISATION: Students formally solve quadratic equations by setting each factor equal to zero (Zero Product Property). They apply this to the Harambee hall problem, obtaining $w = 10$ and $w = -12$, then reason about which solution is physically valid. Multiple Kenyan-context equations are solved in pairs. Students revisit the DQB and refine their model: "A quadratic equation can have two solutions; context tells us which one to use." Lesson 5 — SOLVING BY COMPLETING THE SQUARE: Students are introduced to completing the square as a method for equations that do not factorise easily. Using the Nakuru school garden problem ($x^2 + \dots = 48$), they practise transforming quadratic equations into the form $(x + p)^2 = q$ and solving. The geometric interpretation (literally completing a square diagram) is drawn to

	connect algebra to area. DQB update: "Completing the square always works, even when factorisation fails." Lesson 6 — THE QUADRATIC FORMULA & THE MOMBASA STONE PROBLEM: Students derive the quadratic formula by completing the square on the general form $ax^2 + bx + c = 0$. They apply it to the Mombasa projectile problem ($-5t^2 + 20t + 25 = 0$) and the mandazi pricing problem, comparing results to earlier methods. Students use calculators to verify solutions and discuss the discriminant ($b^2 - 4ac$) to determine the number of real solutions. DQB update: "The quadratic formula is a universal tool — it works for any quadratic equation." Lesson 7 — APPLICATION, MODEL BUILDING & CONSOLIDATION: Students return to the full Harambee hall scenario and present complete solutions using all three methods, justifying their preferred approach. New application problems (land area, fundraising revenue, construction) are solved in groups. Students construct a final cumulative model (poster or digital slide) mapping the journey from recognising quadratic expressions to solving equations in context. The DQB is formally closed — all initial "wonder" questions are answered. Students reflect on how quadratic thinking empowers community problem-solving.
Supporting Phenomena	– A maize farmer in Kericho wants to fence a rectangular plot along one side of a river (so only three sides need fencing). She has exactly 60 metres of wire and wants the enclosed area to be 400 m^2. Students discover that forming the area constraint produces a quadratic equation, and that it may have two possible solutions — only one of which is physically sensible. – A mandazi vendor at Gikomba Market in Nairobi prices her mandazi at Ksh x per piece. She notices that when she raises the price by Ksh 5, the number of mandazis she sells per day drops by 10, yet her total revenue stays at Ksh 2,000. Students form a quadratic equation to find the original price, encountering a real-life revenue model. – A stone is thrown vertically upward from the top of a building in Mombasa. Its height above the ground (in metres) after t seconds is given by $h = -5t^2 + 20t + 25$. Students use the quadratic formula to find when the stone hits the ground, connecting quadratic equations to motion in a physical context they can visualise. – A Nakuru school wants to create a rectangular garden inside a $10 \text{ m} \times 8 \text{ m}$ courtyard, leaving a uniform path of width x metres around it. The garden area must be 48 m^2. Students model the situation as a quadratic equation in x, reinforcing how quadratic equations arise naturally from geometric constraints.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (40 minutes (one lesson; double period optional for DQB)): Predict & Anchor Phenomenon: The Githurai Harambee Hall

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with the Githurai Harambee hall problem, so they feel the limits of arithmetic and linear equations and discover the need for quadratic equations.
Knowledge	Learners will recognise that when an unknown is multiplied by itself ($w \times w$), the equation contains a w^2 term and is quadratic, not linear. They will form $w(w + 2) = 120$, write it as $w^2 + 2w - 120 = 0$, state the general form $ax^2 + bx + c = 0$ (a not equal to 0), and identify $a = 1$, $b = 2$, $c = -120$ for the hall equation.
Skills	Making predictions and testing them with a trial-and-error table ($w = 8$ to 12); modelling a real situation with an equation (NGSS-style SEP: developing and using models, asking questions); comparing linear and quadratic situations (perimeter $4w + 4$ versus area); communicating reasoning in pairs and writing What we notice / What we wonder cards.
Attitudes	Curiosity and perseverance when current tools are not enough; unity and collaboration in pairs and groups; appreciation that mathematics helps communities plan, as in a Harambee.
Key Inquiry Question	How do quadratic expressions and equations differ from linear ones, and why are they necessary? (Also opens: How can we use quadratic equations to model and solve problems in day-to-day activities?)
Purpose in Storyline	Lesson 1 anchors the whole unit. Learners meet the Githurai hall, predict the width, and create a need to know. They end with the equation $w^2 + 2w - 120 = 0$ and an open DQB. Lesson 2 will build on the general form and on $w(w + 2)$ to learn expansion with FOIL. The equation is NOT solved formally here; solving methods come in later lessons.
Safety Notes	Low-risk classroom lesson. Keep aisles clear when groups move to the DQB wall. Use tape or sticky tack for cards, and have learners stand safely when posting. Share manila paper and markers carefully.

B. LESSON OVERVIEW

Learners are shown the Githurai committee's manila-paper sketch: a rectangle labelled width = w , length = $w + 2$, Area = 120 m^2 . They predict the width and explain how they got it. Most guesses come from trial and error. The class tests guesses in a table ($w = 8$ to 12), finds that $w = 10$ gives exactly 120 m^2 but that nobody could be sure without testing, and asks why a linear equation like the perimeter $4w + 4$ cannot give area. Through Think-Pair-Share, learners build $w(w + 2) = 120$ and rewrite it as $w^2 + 2w - 120 = 0$. The teacher introduces the general form $ax^2 + bx + c = 0$ and learners identify $a = 1$, $b = 2$, $c = -120$, then sort expressions as linear or quadratic. The class opens the Driving Question Board with What we notice / What we wonder cards and makes a first model sketch.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Learners view the committee's manila-paper sketch (rectangle,	Display or draw the sketch on the board. Say: "The Githurai	Predict-Share-Reveal (Prediction Tally): Making a prediction before	Look for: (1) learners write a numerical prediction with a reason;

Quadratic systems: a line and a parabola <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Quadratics by factoring <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	width = w, length = $w + 2$, Area = 120 m^2) with the story: the Githurai community is planning a Harambee to raise funds for a new multipurpose hall, and the committee is stuck. Individually, each learner writes in their exercise book a prediction of the width in metres, with one sentence on how they got it. Then they try to write a linear equation for the problem and note why it fails. Learners share in pairs and record their predictions on a class tally on the board.	committee wants a hall floor of exactly 120 square metres, with the length 2 metres more than the width. They tried arithmetic and linear equations and got stuck. Before I teach anything, what do you think the width is?" Give 1 minute of silent writing. WAIT TIME: 10-15 seconds after asking, "What would a linear equation look like here?" Cold-call 4 learners for predictions and record all on the board without saying whether they are right. Ask 2 learners: "How did you get your number?" Say: "Wrong guesses are welcome today; they help us see what tools we need."	instruction activates prior knowledge of area of a rectangle and linear equations, and exposes the limits of those tools. The visible tally of different guesses creates a shared puzzle and a genuine need for new mathematics.	(2) learners link area to length $\times$ width; (3) at least some say they used guess-and-check; (4) learners attempt a linear equation and cannot produce one. Record the range of predictions and any misconceptions (e.g. 120 divided by $2 = 60$, or adding $w + w + 2 = 120$).
Observe Phase VIDEO: Quadratic systems: graphical solution <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Quadratics by factoring (intro) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In pairs, learners test guesses by completing a trial-and-error table for $w = 8, 9, 10, 11, 12$ with columns: width w, length $w + 2$, area $w \times (w + 2)$, and comment. They find: $w = 8, l = 10, 80 \text{ m}^2$ (too small); $w = 9, l = 11, 99 \text{ m}^2$ (too small); $w = 10, l = 12, 120 \text{ m}^2$ (exact); $w = 11, l = 13, 143 \text{ m}^2$ (too big); $w = 12, l = 14, 168 \text{ m}^2$ (too big). They then compute the perimeter $2(w + w + 2) = 4w + 4$ and notice that it is linear and does not give the area. Finally, they notice that w appears multiplied by itself when writing the area.	Hand out the table template or have learners rule it in their books. Circulate and ask: "What happens to the area as w grows from 8 to 12? Does it grow by the same amount each time?" (Differences: 19, 21, 23, 25.) Cold-call 3 pairs to report which row gave exactly 120. Say: "Trial and error worked here, but what if the numbers were not whole? What if the committee had no time to test?" WAIT TIME: 10-15 seconds after asking, "Why can't the perimeter $4w + 4$ tell us the area?" Then ask: "Look at area = $w \times (w + 2)$. What do you notice about w?" Guide towards $w \times w = w^2$.	Pattern Noticing with a Data Table: Learners see the area changing by unequal amounts (19, 21, 23, 25), unlike the steady change of a linear relationship. Comparing the perimeter $4w + 4$ (constant rate of change) with the area builds an evidence-based distinction between linear and non-linear relationships.	Check tables for correct products (80, 99, 120, 143, 168). Look for learners who: identify $w = 10$ as the match; state that the area increases by growing amounts; correctly compute the perimeter $4w + 4$; and say that $w \times w$ gives a squared term. Ask 2 learners to explain in one sentence why the perimeter is linear and the area is not.
Explain Phase	Using Think-Pair-Share, learners	Say: "Let's write down exactly what	Claim-Evidence-Reasoning with	Look for: correct equation $w(w + 2)$

 VIDEO: The quadratic formula <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Linear models word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	write the equation for the hall: $w(w + 2) = 120$. They expand using the distributive law in a simple way ($w \times w + w \times 2$) to write $w^2 + 2w = 120$, then move 120 to the left to get $w^2 + 2w - 120 = 0$. The teacher introduces the general form $ax^2 + bx + c = 0$ (a not equal to 0), and learners label $a = 1$, $b = 2$, $c = -120$. They then classify given expressions and equations as linear or quadratic, e.g. $4w + 4$, $w^2 + 2w - 120$, $3x - 7 = 0$, $2x^2 - 5x + 1 = 0$, and give a reason (highest power 2 versus 1). They explain in their own words why the hall problem needs a quadratic equation.	the committee knows. Area is length times width. Turn to your partner and write an equation using w." WAIT TIME: 10-15 seconds after the question, then cold-call 3 pairs. Write $w(w + 2) = 120$ on the board. Say: "Remove the bracket: w times w is w squared, w times 2 is $2w$." Show $w^2 + 2w - 120 = 0$. Introduce vocabulary: "An expression whose highest power is 2 is a quadratic expression; set equal to something, it is a quadratic equation. The general form is $ax^2 + bx + c = 0$, where a is not zero." Ask: "Why can't a be zero?" (Cold-call 2.) Ask 3 learners to state a, b and c for the hall. Do not solve the equation yet; say: "We will learn how to expand, factorise and solve in the coming lessons. Today we name the problem." Show 6 mixed expressions for quick linear/quadratic sorting using thumbs up or down.	Vocabulary Building: Learners claim the hall equation is quadratic and support it with evidence from the table and the $w \times w$ term. Naming a, b, c connects the concrete hall problem to a general structure, which supports transfer to later contexts.	$= 120$; correct rearrangement to $w^2 + 2w - 120 = 0$; correct $a = 1$, $b = 2$, $c = -120$ (watch for the sign of c); correct linear/quadratic sorting with reasons. Use a 3-question exit check: (1) State a, b, c in $3x^2 - 4x + 5 = 0$. (2) Is $5w + 10$ quadratic? Why? (3) Why is the hall equation not linear?
Driving Question Board (DQB) Creation  VIDEO: Quadratic systems: a line and a parabola <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratic systems <i>Related topic: not an exact match</i>	Groups of 4 write What we notice cards (e.g. area changes by 19, 21, 23, 25; $w \times w$ gives w^2; the perimeter is linear) and What we wonder cards (e.g. Is there a quick way to find w without guessing? Can there be two answers? What if the answer is not a whole number? Can a width be negative? How do we find unknown dimensions for other community plans?). They post cards on the DQB under the driving question: How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems, like building the Harambee hall in Githurai? Learners group similar cards into themes.	Write the driving question on a manila-paper banner at the top of the board or wall. Say: "Each group, write at least two notice cards and two wonder cards. One idea per card." Circulate and prompt: "What surprised you? What would the committee need to know?" WAIT TIME: 10-15 seconds after asking, "Which wonder cards go together?" Cold-call 4 groups to read one card each and place it on the board. Cluster cards into themes such as solving methods, number of solutions, non-whole numbers and other uses. Say: "We will return to this board after every lesson and move cards from wonder to answered."	Driving Question Board (Notice/Wonder Cards): Posting learner-generated questions gives ownership of the learning path and makes curiosity visible. Clustering questions into themes lets learners see how the unit's lessons will answer them.	Check cards for specificity and relevance: do wonder cards ask about solving methods, the number of solutions, or applications? Look for learners who link observations to the table or equation. Note questions that foreshadow later lessons (e.g. the negative width, or non-whole answers) and record them as indicators of readiness.

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar readings				
Model Building Phase  VIDEO: Quadratic systems: graphical solution <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Linear models word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each group starts a cumulative model poster on manila paper titled Githurai Hall Model v1. They draw the hall rectangle with width = w, length = $w + 2$ and Area = 120 m^2. Beside it they write: $w(w + 2) = 120$, $w^2 + 2w - 120 = 0$, general form $ax^2 + bx + c = 0$ with $a = 1$, $b = 2$, $c = -120$. They add a box: Linear: highest power 1 (perimeter $4w + 4$); Quadratic: highest power 2 (area). Below, they write What we still need to find out, with 2 open questions. Learners complete a summary table of what they observed, learned and explained.	Say: "Your model must show the picture, the equation, the general form and one sentence on why this problem is quadratic and not linear." Circulate and check that $c = -120$ has the right sign. WAIT TIME: 10-15 seconds after asking, "What is the one thing our current model cannot do yet?" (Answer: give the width without guessing.) Cold-call 3 groups to share their open questions. Keep posters for revision in each later lesson. Close by saying: "Tomorrow we learn to expand brackets like $(w + 2)(w + 3)$, the first tool for working with quadratics."	Cumulative Model Building: A visual poster that combines diagram, equation and general form helps learners connect representations. Marking the gaps (What we still need) creates a clear target for the next lessons, and the model will be revised in each lesson until Lesson 7.	Check each poster for: correct labelled sketch; $w(w + 2) = 120$; $w^2 + 2w - 120 = 0$; correct a, b, c; a clear linear/quadratic contrast; at least 2 open questions. Look for learners who explain the model to a visitor from another group. Use the summary table to see whether learners can separate what they observed (table, $w \times w$) from what they learned (general form) and explained (why linear fails).

D. TEACHER REFLECTION

Which learners relied only on guess-and-check, and who tried to build an equation? Did learners confuse $w + w + 2 = 120$ or the perimeter with area, and how did I address it? Was the sign of $c = -120$ a common error? Did the wait time (10-15 seconds) give quieter learners a chance to speak, and did I cold-call a balanced mix? Which wonder cards (e.g. negative width, non-whole answers, two solutions) show readiness to be taken up in later lessons, and how will I highlight them on the DQB? What would I change in the time spent on the trial-and-error table versus the DQB creation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe about the Githurai hall problem? Record the trial-and-error table results ($w = 8$ to 12 and areas $80, 99, 120, 143, 168 \text{ m}^2$), the finding that $w = 10$ gives exactly 120 m^2 , that the area grows by different amounts, and that the perimeter $4w + 4$ is linear.
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What did I learn?	What did we learn? Record that when an unknown is multiplied by itself we get a w^2 term and a quadratic equation; the general form $ax^2 + bx + c = 0$ with a not equal to 0; and that for the hall $w^2 + 2w - 120 = 0$, $a = 1$, $b = 2$, $c = -120$.
How does this explain the phenomenon?	How can we explain why the committee got stuck? Write 2-3 sentences using the evidence: area = $w(w + 2)$ has $w \times w$, so it is not linear; the perimeter $4w + 4$ is linear but does not give area; guessing works only if we are lucky, so we need new methods for quadratic equations.

LESSON 2 (40 minutes (one lesson; extend to a double lesson if learners need more practice time)): Expanding and Simplifying Quadratic Expressions: From the Githurai Hall to the Kericho Tea Farm

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners expand a product of two binomials using the distributive law and FOIL, collect like terms to write the result in the form $ax^2 + bx + c$, and explain why an area gives a quadratic expression while a perimeter gives a linear one.
Knowledge	Learners know that $(w + 2)(w + 3) = w^2 + 5w + 6$ and $(2x - 1)(3x + 4) = 6x^2 + 5x - 4$. They know that the area of the Kericho tea-farm plot $(x + 5)(x + 3) = x^2 + 8x + 15$ m ² is quadratic, while its fencing perimeter $4x + 16$ m is linear. They know that expanding always produces the structure $ax^2 + bx + c$, with a, b and c identified.
Skills	Expanding using the distributive law and FOIL; collecting like terms; checking an expansion by substituting a value ($x = 5$ gives 80 m ²); drawing and reading an area model; identifying a, b and c; constructing and explaining a mathematical argument (NGSS-aligned practices: using mathematics, constructing explanations, analysing patterns).
Attitudes	Learners show precision with signs, persevere when a negative term causes an error, respect peers' reasoning in pair work, and appreciate that algebra helps farmers and committees plan.
Key Inquiry Question	How do quadratic expressions and equations differ from linear ones, and why are they necessary? (KICD Key Inquiry 1) Supporting question: Where do quadratic expressions come from in real planning problems, such as the Githurai Harambee hall and a Kericho tea-farm plot?
Purpose in Storyline	In Lesson 1 learners met the Githurai hall expression $w(w + 2)$ and the general form $ax^2 + bx + c$, and saw that the unknown multiplied by itself makes a quadratic. This lesson teaches how a product of two binomials is expanded and simplified into that general form, so that learners see $ax^2 + bx + c$ emerging every time. This prepares for Lesson 3, where they reverse the process by factorising. The DQB is updated with: 'We now know how quadratic expressions are formed.'
Safety Notes	No hazardous materials. Use blunt scissors and paper only when cutting area-model rectangles. Keep walkways clear when learners move to gallery-walk stations. Ensure group work is inclusive so every learner can see the manila paper and area models.

B. LESSON OVERVIEW

Learners begin by recalling the Githurai committee's manila-paper sketch with $w(w + 2) = 120$ and predict what happens when a similar rectangle has both sides written as brackets: a Kericho tea farmer fences a plot of length $(x + 5)$ m and width $(x + 3)$ m. In the Observe Phase, learners build area models with paper rectangles, then expand $(w + 2)(w + 3)$ with the distributive law and FOIL, and compare area (quadratic) with perimeter (linear). In the Explain Phase they expand $(2x - 1)(3x + 4)$, check results by substitution at $x = 5$ (80 m²), and explain how the signs and coefficients behave. They update the DQB with 'What we notice / What we wonder' cards and add a first layer to the cumulative model poster: area of a rectangle with unknown sides leads to $ax^2 + bx + c$.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Combining like terms with distribution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Combining like terms with distribution Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the committee's manila-paper sketch from Lesson 1 (width = w, length = $w + 2$, Area = 120 m^2) and then see a new sketch from a Kericho tea farmer: a rectangular plot with length $(x + 5) \text{ m}$ and width $(x + 3) \text{ m}$. Individually, for 2 minutes, they write on a mini-whiteboard or in their exercise book: (a) 'What will the area expression look like when I multiply $(x + 5)(x + 3)$? Will it have an x^2 term?' (b) 'What will the fencing perimeter look like? Will it have an x^2 term?' They also predict the area if $x = 5$. Then they share with a partner and record one prediction and one reason on a sticky note.	Display both sketches. Say: 'In Lesson 1 we saw that w times $(w + 2)$ gave us a w squared. Today a Kericho tea farmer wants to fence a plot, and both sides have brackets. Before I teach you anything, what do you think the area will look like?' Ask learners to write independently. WAIT TIME: 15 seconds of silence after posing the question before anyone speaks. Cold-call 3 learners (not volunteers) to share predictions, and ask each: 'What makes you think so?' Ask 2 learners to share their prediction for $x = 5$. Record all predictions on the board without judging them as right or wrong. Say: 'Keep your sticky note; we will come back to it at the end.'	Predict-Observe-Explain with a Think-Pair-Share: committing to a prediction first makes learners' current thinking visible (for example, 'just add the brackets') and creates a need for a method. Linking to the Githurai hall keeps the new idea tied to the anchoring phenomenon.	Observe whether learners mention an x^2 term for area and not for perimeter. Look for these indicators: (1) at least 3 of 5 observed pairs write a prediction with a reason; (2) note the number of learners who predict $x^2 + 15$ or $2x + 8$ (a common misconception: multiplying only the first and last terms); (3) tally the predicted values for $x = 5$ on the board. Use the tally to plan which misconceptions to address in the Observe Phase.
Observe Phase  VIDEO: Simplifying polynomials Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Multiply binomials by polynomials Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (area model, about 8 minutes): In pairs, learners use a pre-drawn grid or cut paper pieces to build the rectangle $(w + 2)$ by $(w + 3)$: one big w by w square, 3 strips of w by 1, 2 strips of 1 by w, and 6 unit squares (2 by 3). They count the pieces and write the total area as $w^2 + 3w + 2w + 6$. Part B (about 7 minutes): Using the same pieces, they link each piece to FOIL: First $w \times w = w^2$, Outer $w \times 3 = 3w$, Inner $2 \times w = 2w$, Last $2 \times 3 = 6$. They collect like terms to get $w^2 + 5w + 6$. Part C (about 5 minutes): For the Kericho plot $(x + 5)(x + 3)$, they expand with the distributive law, getting $x^2 + 3x + 5x + 15 = x^2 + 8x + 15 \text{ m}^2$. They then compute the fencing perimeter $2[(x + 5) + (x + 3)] = 4x + 16 \text{ m}$ and record in a table which has an x^2 term and which does not. They test $x = 5$: plot is 10 m by 8 m, area 80 m^2, and $25 + 40 + 15 = 80$; perimeter $4(5) + 16 = 36 \text{ m}$.	Circulate and ask: 'Which piece of your rectangle is the w squared? Show me with your finger.' Say: 'Distributive law means every term in the first bracket multiplies every term in the second bracket. How many multiplications will there be?' (4). Hold up the FOIL card (First, Outer, Inner, Last) and say: 'FOIL is a memory aid for the same four multiplications.' After Part B, pose: 'Why are $3w$ and $2w$ both in the answer, and why can we add them?' WAIT TIME: 10 seconds, then cold-call 3 pairs. After Part C ask: 'The farmer buys wire for the fence. Which expression does she use, area or perimeter? Why is the perimeter linear while the area is quadratic?' WAIT TIME: 10 seconds, cold-call 2 learners. Redirect a pair that wrote $x^2 + 15$: 'Point to the $3x$ and $5x$ pieces in your model. Where did they go in your answer?'	Concrete-Representational-Abstract progression using an area model: the physical rectangle pieces show why four products appear and why like terms combine. Comparing area and perimeter side by side (Contrasting Cases) builds understanding of why quadratic expressions differ from linear ones.	Check each pair's recorded steps. Indicators of understanding: (1) the pair identifies all four products in $(w + 2)(w + 3)$ and writes $w^2 + 3w + 2w + 6$ before simplifying; (2) correctly combines $3w + 2w = 5w$; (3) writes perimeter as $4x + 16$ with no x^2 term; (4) verifies the Kericho area at $x = 5$ as 80 m^2. Use a quick thumbs-up/sideways/down check: 'Can you explain why $3w$ and $2w$ are like terms?' Note pairs for follow-up if fewer than 4 of 6 pairs show all four indicators.

Explain Phase  VIDEO: Simplifying polynomials - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Combining like terms with distribution Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in groups of 3-4 to expand $(2x - 1)(3x + 4)$ without a physical model, using FOIL: First $2x \times 3x = 6x^2$, Outer $2x \times 4 = 8x$, Inner $-1 \times 3x = -3x$, Last $-1 \times 4 = -4$. They collect like terms to get $6x^2 + 5x - 4$ and identify $a = 6$, $b = 5$, $c = -4$. Each group then writes a 'Why does it work?' explanation card comparing the signs in $(2x - 1)(3x + 4)$ with those in $(w + 2)(w + 3)$, checks the result by substituting $x = 1$ into both forms $(1)(7) = 7$ and $6 + 5 - 4 = 7$, and finally works two more expansions of their own choice and swaps with another group to check. They also revisit the hall: expand $w(w + 2) = w^2 + 2w$, connecting to the Lesson 1 form where $a = 1$, $b = 2$ before subtracting 120, and note that the structure $ax^2 + bx + c$ appears every time. Finally they return to the Predict sticky notes and revise their prediction with evidence.	Say: 'You have used FOIL on positive numbers. What changes when a bracket has a minus sign?' WAIT TIME: 15 seconds, then cold-call 2 groups. Listen for 'the sign stays with the term'. Prompt: 'Inner is -1 times 3x. Is that +3x or -3x? How do you know?' Display a common error, $6x^2 + 11x - 4$, and ask: 'Where might this have come from? Find the mistake.' ($8x + 3x$ instead of $8x - 3x$). WAIT TIME: 10 seconds, cold-call 3 learners. Then ask: 'In all of our examples, what shape does the answer have?' Guide to $ax^2 + bx + c$. Ask: 'How is the Kericho area $x^2 + 8x + 15$ like the hall expression we met in Lesson 1?' Use a gallery walk of the groups' explanation cards for 4 minutes and ask each learner to leave one 'I agree because...' note. Return to the predictions: 'Whose prediction was close? What did we learn that changes your prediction?'	Error Analysis and Claim-Evidence-Reasoning: learners analyse a worked error, then write a claim (the expansion), evidence (substitution check), and reasoning (distributive law). Generalising across several examples helps them see the consistent structure $ax^2 + bx + c$, so expansion is understood and not just memorised.	Collect each group's explanation card. Indicators: (1) correct expansion $6x^2 + 5x - 4$; (2) correct identification of $a = 6$, $b = 5$, $c = -4$, with c negative; (3) a substitution check that gives equal values on both sides; (4) a written statement that the sign belongs to the term that follows it; (5) correct peer-checking of at least 1 of 2 self-chosen expansions. Use a 3-question exit ticket: expand $(x + 4)(x + 1)$; expand $(x - 2)(x + 5)$; state a, b, c for your answer to the second question. Success = 2 of 3 correct (expected answers $x^2 + 5x + 4$; $x^2 + 3x - 10$ with $a = 1$, $b = 3$, $c = -10$).
Driving Question Board (DQB) Creation  VIDEO: Multiplying binomials by polynomials: area model Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Multiply binomials by polynomials Source: Khan Academy (English - US curriculum)  Search ARES for similar	Learners return to the class Driving Question Board: 'How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real planning problems — like building the Harambee hall in Githurai?' In pairs, they write one 'What we notice' card (for example, 'Area gives an x^2 term, perimeter does not') and one 'What we wonder' card (for example, 'If we can expand to get $x^2 + 2x - 120$, can we go backwards to find w?'). They post cards under the headings. The class also adds the statement on the board: 'We now know how quadratic expressions are formed.' They then sort the cards to see which questions can be answered in the next lessons.	Say: 'Our Githurai committee is still stuck. What have we learned today that helps them, and what do we still not know?' Give each pair 2 cards and 2 minutes to write. WAIT TIME: 10 seconds before collecting. Cold-call 4 pairs to read a card aloud and to say where to place it on the board. Point to a wonder card and say: 'If expanding turns $(w + 2)(w + 3)$ into $w^2 + 5w + 6$, what would going backwards be called? We will explore that in the next lesson.' Do not answer wonder cards about solving the equation yet. Write on the board: 'We now know how quadratic expressions are formed.'	Driving Question Board with 'What we notice / What we wonder' cards: visible tracking of growing understanding keeps the sequence coherent and lets learners' own questions (such as factorising and solving) motivate the next lessons.	Read the cards. Indicators: (1) at least 80% of notice cards mention area/perimeter, x^2, or the pattern $ax^2 + bx + c$; (2) at least 3 wonder cards ask about reversing expansion or solving for w; (3) cards use correct vocabulary (expand, like terms, coefficient). If a card states 'perimeter has x^2', discuss it at the start of Lesson 3.

readings				
Model Building Phase  VIDEO: Combining like terms with distribution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Combining like terms with negative coefficients <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In their groups, learners add to the cumulative model poster (started in Lesson 1). Today's panel is titled 'How a quadratic expression is formed'. They draw (1) a rectangle with sides $(x + 5)$ and $(x + 3)$ and its area model split into x^2, $5x$, $3x$ and 15; (2) the expansion steps with FOIL labels; (3) the result $x^2 + 8x + 15$ with a, b, c labelled ($a = 1$, $b = 8$, $c = 15$); (4) the perimeter $4x + 16$ shown as linear; and (5) a connection arrow to the Githurai hall: $w(w + 2) = w^2 + 2w$, which becomes $w^2 + 2w - 120 = 0$ once the area 120 m^2 is subtracted from both sides. Under 'What we still need', they write: 'How do we go backwards from $x^2 + bx + c$ to the brackets?' Each group member explains one part of the panel to a partner.	Say: 'Your model has to show how the expression is formed, not just the answer. Where in your drawing does the $8x$ come from?' WAIT TIME: 10 seconds, then cold-call 3 groups to explain one part of their poster. Prompt: 'How does the Kericho plot help the Githurai committee? What is the same about the two situations?' (Both have an area made by multiplying two lengths that include an unknown.) Ask: 'What do we still need to be able to find $w = 10$?' Collect responses and say: 'Next lesson we reverse the process.' Close by returning the sticky notes and asking each learner to write one sentence beginning 'I used to think ... now I think ...'.	Cumulative Model Building: learners combine drawings, symbols and words to show how the area of a rectangle with unknown sides produces $ax^2 + bx + c$. Explaining their model to a peer consolidates understanding, and identifying gaps ('What we still need') links to the next lesson.	Check each poster panel. Indicators: (1) area model has the four parts correctly labelled x^2, $5x$, $3x$, 15; (2) expansion steps match the diagram; (3) a, b, c correctly labelled (1, 8, 15); (4) perimeter $4x + 16$ correctly contrasted as linear; (5) a clear link to $w(w + 2)$ from the Githurai hall; (6) a gap statement about reversing expansion. Success = at least 5 of 6 indicators for each group.

D. TEACHER REFLECTION

After the lesson, reflect: Which learners confused 'adding' brackets with 'multiplying' them (for example, writing $x^2 + 15$)? Did the area model help them to see why four products are needed? Which learners made sign errors with negative terms such as $(2x - 1)$, and did the error-analysis activity help? Did the 15-second wait times give quieter learners a chance to contribute, and did cold-calling feel safe? How well did learners connect the Kericho plot back to the Githurai hall? Note which learners are ready to factorise in Lesson 3 and who needs more practice with FOIL and like terms. Plan one extra mini-practice with two positive and two negative-sign expansions for those who need it.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that multiplying two brackets such as $(w + 2)(w + 3)$ gives four products. In the area model of the Kericho plot $(x + 5)(x + 3)$, the pieces were x^2 , $5x$, $3x$ and 15 . The area $x^2 + 8x + 15 \text{ m}^2$ has an x^2 term, but the fencing perimeter $4x + 16 \text{ m}$ does not. At $x = 5$, the plot is 10 m by 8 m , with area 80 m^2 and perimeter 36 m .
What did I learn?	We learned to expand a product of two binomials using the distributive law and FOIL (First, Outer, Inner, Last), then collect like terms. For example, $(w + 2)(w + 3) = w^2 + 5w + 6$ and $(2x - 1)(3x + 4) = 6x^2 + 5x - 4$. The answer always has the form $ax^2 + bx + c$, and signs must stay with the terms that follow them.

How does this explain the phenomenon?

We can explain that the Githurai hall expression $w(w + 2)$ and the Kericho plot area $(x + 5)(x + 3)$ are quadratic because an unknown length is multiplied by another length that contains the unknown, creating an x^2 term. A perimeter adds lengths, so it stays linear. We now know how quadratic expressions are formed, but we still need a way to go backwards to find the brackets and later the width w of the hall.

LESSON 3 (40 minutes (1 double lesson of 2 x 40 minutes recommended if time allows; core plan fits 80 minutes)): Factorisation of Quadratic Expressions: Undoing Expansion to Find the Hidden Hall Dimensions

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners reverse the expansion process from Lesson 2 to factorise quadratic expressions of the form $x^2 + bx + c$ and $ax^2 + bx + c$, including perfect square trinomials and the difference of two squares, and use this to write the Githurai hall expression $w^2 + 2w - 120$ as $(w + 12)(w - 10)$.
Knowledge	Factorise means write an expression as a product of brackets. It is the reverse of expansion. For $x^2 + bx + c$ we look for two numbers whose product is c and whose sum is b . Perfect square: $x^2 + 6x + 9 = (x + 3)^2$. Difference of two squares: $x^2 - 49 = (x - 7)(x + 7)$. For $ax^2 + bx + c$, such as $2x^2 + 7x + 3 = (2x + 1)(x + 3)$, we split the middle term or use trial brackets and check by expanding. Hall: $w^2 + 2w - 120 = (w + 12)(w - 10)$ using the factor pair 12 and -10 (product -120, sum 2). Gikomba mandazi: $x^2 + 7x - 30 = (x + 10)(x - 3)$.
Skills	Analysing and interpreting data (finding factor pairs from a table of product and sum); using mathematics and computational thinking (reversing FOIL); constructing explanations (justifying a factorisation by expanding to check); communication and collaboration in pairs and groups.
Attitudes	Unity: learners work in pairs and groups to find and check factor pairs. Responsibility: learners check each other's factorisations by expanding. Appreciation that mathematics helps the community plan, as the Githurai committee does. Life skills (self-esteem): every learner contributes a factor pair or a check.
Key Inquiry Question	How do quadratic expressions and equations differ from linear ones, and why are they necessary? (KICD Key Inquiry 1) Supporting question for this lesson: How can we 'undo' expansion to find the hidden brackets inside $w^2 + 2w - 120$?
Purpose in Storyline	Lesson 1 showed that the hall gives $w(w + 2) = 120$, i.e. $w^2 + 2w - 120 = 0$, which is a quadratic equation different from a linear one. Lesson 2 taught expanding binomials with the distributive law and FOIL to get $ax^2 + bx + c$. Lesson 3 uses that structure in reverse: learners factorise the hall expression into $(w + 12)(w - 10)$ and meet the Gikomba mandazi revenue expression as a new factorisation challenge. Lesson 4 will then set each factor equal to zero (Zero Product Property) to find the hall's width. This lesson stops at the factorised form and does NOT yet solve for w .
Safety Notes	No hazardous materials. Use paper tiles, manila paper and markers. Keep walkways clear when learners move to the gallery wall. Pair learners so that everyone participates and no one is left out. Scissors, if used to cut algebra tiles from paper, should be blunt-ended and handled with care.

B. LESSON OVERVIEW

The Githurai committee's manila-paper sketch gives the hall expression $w^2 + 2w - 120$, but they cannot yet see the width inside it. In this lesson learners first predict how to get back from $w^2 + 2w - 120$ to two brackets. They then gather evidence in the Observe phase by reversing FOIL on simple expressions ($x^2 + 6x + 9$, $x^2 - 49$, $2x^2 + 7x + 3$), using a product-and-sum factor-pair table and a paper area-tile model. In the Explain phase they factorise the hall expression to $(w + 12)(w - 10)$ and tackle the Gikomba mandazi revenue $x^2 + 7x - 30 = (x + 10)(x - 3)$ as a challenge. They update the Driving Question Board with 'Factorisation lets us undo expansion to find hidden values' and revise the cumulative model poster. Solving for w (and rejecting -12) is left to Lesson 4.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Factoring Perfect Square Trinomials: A Sample Application Source: CK-12  Search ARES for similar videos  HTML: Factorization using Perfect Square Trinomials (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look at the committee's manila-paper sketch again (width = w , length = $w + 2$, Area = 120 m ²) and the expression $w^2 + 2w - 120$ from Lesson 1. On a sticky note or in their exercise book they answer individually: 'Lesson 2 turned two brackets into one expression. If we start with $w^2 + 2w - 120$, how could we get back to two brackets? What do you think the two brackets might look like?' They write one prediction and one reason, e.g. 'I think the brackets are $(w + \text{something})(w - \text{something})$ because the middle term is small'. They then share with a partner.	Display the sketch on manila paper or the board. Say: 'Yesterday we expanded $(w + 2)(w + 3)$ to get $w^2 + 5w + 6$. Today the committee hands us the answer and asks for the brackets. Write your prediction silently.' Give WAIT TIME of 10-15 seconds after asking: 'What could the brackets look like?' Give 2 minutes of silent writing, then 1 minute of pair talk. Cold-call 4 learners (not volunteers) with: 'Read your prediction and say why.' Record all predictions on the board without judging them as right or wrong. Ask: 'What would we multiply to check a prediction?' and let learners answer that expanding is the check.	Predict-Share-Revisit: learners commit to a prediction and a reason before instruction, so that later evidence can confirm or revise their thinking. Naming the check (expanding with FOIL) links Lesson 2 to this lesson and establishes the idea that factorising is the reverse of expanding.	Observe whether learners: (1) write two brackets of the form $(w + _)(w + _)$ in their prediction; (2) mention that the constant 120 or the middle term 2 must be used; (3) say that expanding checks the answer. Tally on a clipboard how many predictions use a negative sign. Learners who write only a number for w (e.g. ' $w = 10$ ') are still thinking by trial-and-error as in Lesson 1; note them for support in the Observe phase.
Observe Phase  VIDEO: Factoring difference of squares: two variables (example 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Factorization using Perfect Square Trinomials Practice Source: CK-12  Search ARES for similar readings	Learners work in pairs through three evidence stations on worksheets and with paper area tiles. Station 1 ($x^2 + bx + c$): they reverse FOIL on $x^2 + 6x + 9$. They fill a product-and-sum table: pairs with product 9 are (1, 9), (3, 3), (-1, -9), (-3, -3); the pair with sum 6 is (3, 3), giving $(x + 3)(x + 3) = (x + 3)^2$, a perfect square trinomial. They expand $(x + 3)^2$ to check. Station 2 (difference of two squares): they expand $(x - 7)(x + 7)$ and see that the middle terms $-7x$ and $+7x$ cancel, giving $x^2 - 49$; they then factorise $x^2 - 49 = (x - 7)(x + 7)$ and note that $49 = 7^2$ and there is no x term. Station 3 ($ax^2 + bx + c$): for $2x^2 + 7x + 3$ they find a x $c = 6$ and look for two numbers with product 6 and sum 7 (1 and 6), split the middle term: $2x^2 + x + 6x + 3$, group: $x(2x + 1)$	Set up three stations on desks or as three worksheet sections. Say: 'Your job is to be detectives. Look for a pattern between the numbers in the brackets and the numbers in the expression.' Circulate and ask each pair: 'Which two numbers multiply to give the constant? Which two add to give the middle coefficient?' Give WAIT TIME of 10-15 seconds before any hint. Prompt: 'Why did the x terms disappear in $x^2 - 49$?' After 15 minutes, stop the class and cold-call 3 pairs, one per station: 'What pattern did you notice?' Write the three patterns on the board: product = c and sum = b ; perfect square $(x + 3)^2$; difference of two squares $(x - 7)(x + 7)$. Ask the class: 'Where did you see that same expansion in Lesson 2?' For learners who struggle with $2x^2 +$	Pattern Seeking with Reversal: learners compare the forward process (FOIL from Lesson 2) with the backward process (factorising) across three cases and generalise from the data. The factor-pair table makes the thinking visible, and the area-tile rectangle gives a concrete picture of why a perfect square has two equal sides, linking back to the rectangular hall.	Look for: (1) learners list factor pairs systematically, including negative pairs, in a table; (2) learners write $(x + 3)(x + 3)$ as $(x + 3)^2$; (3) learners state that $x^2 - 49$ has no middle term because the two middle terms cancel; (4) learners check at least one factorisation by FOIL. Exit signal: ask for a thumbs up, sideways or down on 'I can find two numbers with a given product and sum'. Pairs who cannot complete the table for $2x^2 + 7x + 3$ get a short mini-conference with the teacher during the Explain phase.

	+ 3(2x + 1) = (2x + 1)(x + 3), and check by FOIL. Learners record each factor-pair table in their books and use paper tiles (x^2 tile, x strips, unit squares) to build a rectangle for $x^2 + 6x + 9$ with sides (x + 3) and (x + 3). Each pair records: 'What we observed about the numbers in the brackets and the numbers in the expression'.	$7x + 3$, say: 'Multiply a and c first: $2 \times 3 = 6$. Which two numbers make 6 and add to 7?' Remind learners that every factorisation is checked by expanding.		
Explain Phase  VIDEO: Factoring Perfect Square Trinomials: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Difference of squares intro Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A, the hall: Learners return to the committee's expression $w^2 + 2w - 120$. In pairs they build a product-and-sum table for 120 with the constant negative: factor pairs of -120 include (10, -12), (-10, 12), (8, -15), (-8, 15), (6, -20), (-6, 20) and so on, and they look for sum +2. They find 12 and -10 (product -120, sum 2) and write $w^2 + 2w - 120 = (w + 12)(w - 10)$. They expand with FOIL to check: $w^2 - 10w + 12w - 120 = w^2 + 2w - 120$. They explain in a sentence why one factor is positive and one negative. They notice that (w + 12) (w - 10) contains the numbers 10 and 12, the same numbers they saw in the trial-and-error table in Lesson 1, but they do NOT solve yet; they write 'What we wonder: How do the brackets give us the width?' Part B, the Gikomba challenge: A mandazi vendor in Gikomba, Nairobi, has (x + 3) customers each paying (x + 4) shillings, so revenue $R = (x + 3)(x + 4) = x^2 + 7x + 12$ KES. The vendor wants $R = 42$ KES, which gives $x^2 + 7x - 30$ (after moving 42 across). In groups of four, learners factorise $x^2 + 7x - 30$ by finding two numbers with product -30 and sum 7 (10 and -3), writing (x + 10)(x - 3). They check by expanding: $x^2 - 3x + 10x - 30 = x^2 + 7x - 30$. Each group	Say: 'The committee gave us $w^2 + 2w - 120$. Can we now find the hidden brackets? Use your table.' Give WAIT TIME of 10-15 seconds before taking any answer. Cold-call 3 pairs: 'Which factor pair did you choose, and how do you know the sum is 2?' Ask the class: 'Why can't 8 and -15 work?' (sum is -7) Write the factorisation on the board and have learners expand it aloud in chorus to check. Then say: 'Now the Gikomba mandazi vendor needs help. Same method, new numbers.' Circulate, asking groups: 'What is your product? What is your sum? How can you test it quickly?' Cold-call 4 groups to share their factor pair and check. Address misconceptions explicitly: (1) writing (x - 10)(x + 3) instead of (x + 10)(x - 3): ask 'Expand it. Does the middle term come out +7x or -7x?'; (2) forgetting that the sign of the larger number matches the sign of b. State clearly: 'Today we only factorise. We do not yet find w or x; that is Lesson 4.' End with: 'What is the same about the hall and the mandazi problem?' and take 2 responses.	Claim-Evidence-Reasoning with Reversal Check: learners state a claim (the factorisation), give evidence (the factor-pair table and the FOIL expansion), and give reasoning (factorising is expanding in reverse, so expanding the answer must return the original expression). Applying the same method to the hall and then to a new context (Gikomba mandazi revenue) helps learners transfer the idea rather than memorise one example.	Look for: (1) the factor pair 12 and -10 chosen and justified using both product -120 and sum 2; (2) (w + 12)(w - 10) written and checked by expanding; (3) (x + 10)(x - 3) written for the mandazi expression with a correct check; (4) learners resisting the urge to solve for w and instead writing a wonder question. Quick check on mini whiteboards: 'Factorise $x^2 + 5x + 6$' (answer (x + 2)(x + 3)); count correct responses. Aim for at least 80% correct; if lower, reteach with the product-and-sum table before the next lesson. Collect pair work for the three tasks to identify learners needing extra support.

	completes the sentence: 'The steps I follow to factorise $x^2 + bx + c$ are...' Groups then write a short comparison: how is this expression like the hall's expression, and how is it different? (Both have a negative constant; the numbers differ.) They leave the solving, and the rejection of -10, for Lesson 4.			
Driving Question Board (DQB) Creation  VIDEO: Factoring difference of squares: two variables (example 2) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Difference of squares Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners add to the class Driving Question Board using 'What we notice / What we wonder' cards. Each pair writes one 'What we notice' card, for example: 'We notice that $w^2 + 2w - 120$ can be written as $(w + 12)(w - 10)$' or 'We notice that the numbers 12 and -10 multiply to -120 and add to 2'. Each pair also writes one 'What we wonder' card, for example: 'We wonder how $(w + 12)(w - 10)$ gives the width of the hall' or 'We wonder why one number is negative and what a negative width would mean'. They place the cards on the DQB under the driving question about the Harambee hall in Githurai. The class then adds a DQB update statement: 'Factorisation lets us undo expansion to find hidden values.' Learners also note the Gikomba mandazi problem on a card: 'We wonder what x is when revenue is 42 KES'.	Hand out cards and say: 'Write what you noticed today and what you still wonder about the hall.' Allow 3 minutes of silent writing. Give WAIT TIME of 10-15 seconds after asking: 'Which wondering cards connect to each other?' Cold-call 4 pairs to read a card and place it on the DQB. Cluster cards into themes such as 'finding the brackets', 'negative numbers', 'getting the width' and 'checking answers'. Point to the 'Factorisation lets us undo expansion to find hidden values' statement and say: 'This is our new piece of evidence. We still do not know the width. What step comes next?' Guide learners to say that each bracket might be set equal to zero, without teaching it, and tell them that Lesson 4 will investigate this. Ask: 'Which wonder cards did today's lesson answer, and which are new?' and move answered cards to a 'We have learned' strip.	Question Clustering on the DQB: learners link new observations to the unresolved driving question, track what is answered and unanswered, and see the lesson as one step in a longer investigation. This keeps the Githurai hall as the thread that connects every lesson.	Check that at least 80% of pairs submit one 'notice' and one 'wonder' card. Look for notice cards that mention the factor pair (12 and -10) or the reverse of expansion, and wonder cards that show a genuine next step (e.g. how to get w from the brackets). Note any cards that show a misconception, such as '$w = 12$ because of the 12 in the bracket', and address them at the start of Lesson 4.
Model Building Phase  VIDEO: Factoring difference of squares: shared factors <i>Related topic: not</i>	Learners update the cumulative model poster for the Harambee hall, which in earlier lessons showed the hall sketch, the equation $w(w + 2) = 120$ and the expanded form $w^2 + 2w - 120 = 0$. In groups of four they add a new panel titled 'Factorising: undoing expansion'. The panel contains: (1)	Say: 'Our model poster must show how we got from the sketch to the brackets. Add a panel that someone from another school could follow.' Provide manila paper and markers (in the same spirit as the committee's sketch). Give WAIT TIME of 10-15 seconds after asking: 'How is factorising linked to	Cumulative Model Revision: learners extend a visible, shared model of the hall problem by adding the new relationship (expanding and factorising as inverse processes). Representing the idea as a diagram, a table and worked examples builds a multi-representation understanding and	Use a checklist for each group poster: (1) two-way expand/factorise diagram present and correct; (2) factor pair 12 and -10 shown with product -120 and sum 2; (3) at least one perfect square or difference of two squares example correct; (4) Gikomba example $(x + 10)(x - 3)$

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Perfect squares Source: Khan Academy (English - US curriculum) Search ARES for similar readings	a two-way arrow diagram, with 'expand (FOIL)' going from $(w + 12)(w - 10)$ to $w^2 + 2w - 120$ and 'factorise' going back; (2) the factor-pair table for 120 with the pair 12 and -10 circled; (3) the three patterns from the Observe phase, $x^2 + 6x + 9 = (x + 3)^2$, $x^2 - 49 = (x - 7)(x + 7)$ and $2x^2 + 7x + 3 = (2x + 1)(x + 3)$; (4) the Gikomba mandazi example $x^2 + 7x - 30 = (x + 10)(x - 3)$. Each group writes one sentence at the bottom: 'Factorising helps the Githurai committee because...' and adds a placeholder box 'Next: use the brackets to find w' for Lesson 4. Finally, learners complete the three-column summary table (see below) in their books.	expanding in the diagram?' Circulate and ask each group: 'Where does the 12 come from? Why is -10 negative?' Cold-call 3 groups to present their arrow diagram in 30 seconds each; ask the class to give one 'star' (what is clear) and one 'step' (what could be clearer). Remind learners: 'We have not yet found the width. Write that on your poster as the next step.' Close with a whole-class check: 'Hands up if you can factorise $w^2 + 2w - 120$ without notes.' Collect the summary tables as an exit ticket.	records the progress of the class thinking across lessons.	correct; (5) next-step placeholder for finding w. Exit ticket: 'Factorise $x^2 + 7x + 12$ and check by expanding' (answer $(x + 3)(x + 4)$, the Gikomba revenue before the target is applied). Learners who miss the check step are given extra practice at the start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Did every learner use the product-and-sum table, or did some still rely on trial-and-error guessing as in the Lesson 1 predictions? (2) Which sign errors appeared most often (for example $(x - 10)(x + 3)$ written for $(x + 10)(x - 3)$), and did expanding help learners find and fix them? (3) Could learners explain why factorising is the reverse of expanding in their own words, using the hall as an example? (4) Did anyone start solving for w too early, and what does this tell me about readiness for Lesson 4? (5) Which cards on the DQB show the most curiosity, and how can I use them to open the next lesson? (6) How well did the Gikomba mandazi context hold learner interest compared with the hall? Note which learners need more practice with $ax^2 + bx + c$, and plan a short warm-up using $2x^2 + 7x + 3$ at the start of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Example: When we reversed FOIL, the two numbers in the brackets multiplied to give the constant and added to give the middle coefficient. $x^2 + 6x + 9$ became $(x + 3)^2$, $x^2 - 49$ became $(x - 7)(x + 7)$, and $2x^2 + 7x + 3$ became $(2x + 1)(x + 3)$.
What did I learn?	What did we learn? Example: Factorising means writing an expression as a product of brackets, and it is the reverse of expanding. We can always check a factorisation by expanding it. Perfect squares and the difference of two squares are special patterns.
How does this explain the phenomenon?	How does this help explain the Githurai hall? Example: The hall expression $w^2 + 2w - 120$ factorises as $(w + 12)(w - 10)$ using the factor pair 12 and -10 (product -120, sum 2). The same method works for the Gikomba mandazi revenue $x^2 + 7x - 30 = (x + 10)(x - 3)$. We have the hidden brackets, but we still wonder how they give the width; that is the next step.

LESSON 4 (40 minutes (one lesson; Predict 5 min, Observe 12 min, Explain 12 min, DQB 5 min, Model Building 6 min)): Solving Quadratic Equations by Factorisation: Zero Product Property and the Githurai Harambee Hall

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners solve quadratic equations by factorisation using the Zero Product Property, then use the context to choose the valid solution, so that the Githurai committee can state the hall dimensions.
Knowledge	By the end of the lesson the learner should be able to: (a) state the Zero Product Property: if $A \times B = 0$ then $A = 0$ or $B = 0$; (b) solve a quadratic equation in general form $ax^2 + bx + c = 0$ by factorising and setting each factor equal to zero; (c) obtain the two solutions $w = 10$ and $w = -12$ for the hall equation $(w + 12)(w - 10) = 0$; (d) check each solution in the context and reject the impossible one, giving width 10 m, length 12 m; (e) solve the Kisumu fish pond equation $w^2 + 3w - 40 = 0$ to get $w = 5$ m, length 8 m.
Skills	Developing and using models (writing and solving an equation from the sketch); Analysing and interpreting data (comparing the trial-and-error table with the algebraic solutions); Constructing explanations (justifying why a solution is rejected); Engaging in argument from evidence (defending which solution is valid); Using mathematics and computational thinking (substitution checks). Core competencies: Critical Thinking and Problem Solving; Communication and Collaboration.
Attitudes	Learners show Unity by working in pairs and sharing roles; Responsibility by checking every answer in context; Integrity by honestly showing every step of working; and they appreciate the use of quadratic equations in solving community planning problems.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities? (supporting: How do quadratic equations differ from linear ones, since a linear equation gives one solution and a quadratic can give two?)
Purpose in Storyline	Lessons 1 to 3 gave learners the equation $w^2 + 2w - 120 = 0$ (Lesson 1) and the skill of factorising it into $(w + 12)(w - 10)$ (Lesson 3). Lesson 4 is the first time the committee actually gets the dimensions: the Zero Product Property turns the factorised equation into two simple linear equations. Learners discover that a quadratic can have two solutions and that context decides which one is used. This sets up Lesson 5 (completing the square for an equation that does not factorise).
Safety Notes	No hazardous materials. Pairs work at desks with mini-whiteboards or exercise books; keep aisles clear when learners move to the DQB wall. Manila paper and markers are used for the cumulative model poster; use markers with caps on and keep them away from clothes and faces. Be sensitive to learners who feel anxious about wrong answers: normalise errors as evidence.

B. LESSON OVERVIEW

The Githurai committee's manila-paper sketch (width = w , length = $w + 2$, Area = 120 m²) is displayed again. In earlier lessons learners wrote $w(w + 2) = 120$ as $w^2 + 2w - 120 = 0$ and factorised it as $(w + 12)(w - 10) = 0$. Learners first predict what values of w could make that product equal to zero. They then investigate the Zero Product Property with number puzzles (e.g. what two numbers multiply to give 0?), apply it to the hall to get $w = 10$ or $w = -12$, and test both solutions against the trial-and-error table ($w = 10$ gives $10 \times 12 = 120$ m²). They reason that a width cannot be negative, so the hall is width 10 m, length 12 m. In pairs they then solve the Kisumu fish pond problem ($w^2 + 3w - 40 = 0$, $(w + 8)(w - 5) = 0$, $w = 5$ m, length 8 m, $5 \times 8 = 40$). The class updates the DQB and revises the cumulative model with the idea: a quadratic equation can have two solutions; context tells us which one to use.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the committee's manila-paper sketch and the factorised equation $(w + 12)(w - 10) = 0$ written on the board. Individually, on a mini-whiteboard or in their books, they write: (1) one value of w that they think makes the left side equal to 0, and (2) a reason. They then share with a partner. Typical predictions include $w = 10$, $w = 12$, $w = -12$, or 'it must be 120'. Learners also predict: 'Will there be one answer or more than one?' This links directly to the Harambee hall: the committee needs the width.	Display the sketch and write $(w + 12)(w - 10) = 0$. Say: "Last lesson we factorised the hall equation. Today the committee wants a number for w . Write down one value of w that you think makes this whole product equal to zero, and write one sentence on why." Give 1 minute of silent writing, then WAIT TIME of 10 seconds before any sharing. Say: "Turn to your partner and compare for 1 minute." Cold-call 3 pairs: "What did your partner predict, and why?" Record every prediction, including wrong ones, on the board under 'Our predictions' without judging. Ask: "Do you think there will be one answer or two? Show with fingers." Tally the hands. Say: "Keep your prediction. We will test it with evidence."	Predict-Share-Record (eliciting prior ideas): Writing a prediction before instruction exposes learners' existing ideas (such as 'the answer is 120' or ' w must be 12'), and recording them publicly creates a shared set of ideas to test. This builds a need to know the rule that governs a product equal to zero.	Listen for and note: (1) how many learners write a value that actually makes a bracket zero (10 or -12) versus those who give 120 or 12 without a reason; (2) whether learners mention that 'one of the brackets must be zero'; (3) the count of learners predicting one answer versus two. Check on 5 mini-whiteboards during circulation. Indicator of readiness: at least half the class gives a value linked to a bracket. If fewer, spend extra time on the number puzzles in the Observe Phase.
Observe Phase  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  EXERCISE: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (4 min): In pairs, learners do a 'Product Puzzle' on the board. They find pairs of numbers A and B so that $A \times B = 0$, e.g. 0×7 , 5×0 , $(-3) \times 0$. They try to find $A \times B = 0$ where neither A nor B is 0 and report that they cannot. They state the Zero Product Property: if $A \times B = 0$ then $A = 0$ or $B = 0$. Part B (4 min): Applying it to the hall, they write $w + 12 = 0$ or $w - 10 = 0$, solve each to get $w = -12$ or $w = 10$, and substitute both in $(w + 12)(w - 10)$ to confirm each gives 0. They compare with the trial-and-error table from Lesson 1 ($w = 8$, area 80; $w = 9$, area 99; $w = 10$, area 120; $w = 11$, area 143; $w = 12$, area 168) and notice $w = 10$ is the exact row. Part C (4 min): They test $w = -12$: length would be $-12 + 2 = -10$ m, so the 'area' $(-12)(-10) = 120$ is arithmetic only, but a hall	Part A: Write on the board " $A \times B = 0$ " and say: "With your partner, find as many pairs of numbers as you can. Then try to find a pair where NEITHER number is zero. You have 2 minutes." WAIT TIME of 10 seconds after the question before circulating. Ask 3 cold-called pairs: "Can you find a pair where neither is zero? What does that tell us?" Write the Zero Product Property on the board and have the class say it together. Part B: Say: "The committee's equation is $(w + 12)(w - 10) = 0$. Which two things could be zero? Write both as little equations and solve them." Circulate and ask: "Why do we set EACH bracket equal to zero and not equal to 120?" Cold-call 2 learners to show substitution of $w = 10$: $(22)(0) = 0$ and $w = -12$: $(0)(-22) = 0$. Part C: Point to the	Pattern-seeking with number puzzles followed by Claim-Evidence-Reasoning: Learners generate the Zero Product Property from examples (evidence) before it is named, then make a claim about the hall's solutions and support it with substitution (evidence) and the Lesson 1 table. This makes the rule meaningful rather than a memorised step, and links the algebra back to the trial-and-error data.	Observable indicators: (1) learners correctly write both $w + 12 = 0$ and $w - 10 = 0$ (not $w + 12 = 120$); (2) learners correctly solve to $w = -12$ and $w = 10$; (3) learners substitute both values back and show a zero product; (4) learners match $w = 10$ to the 10 m x 12 m row in the table. Note any learner who writes $(w + 12) = 0$ and $(w - 10) = 120$ and address it using a one-minute mini-conference. Collect 5 pairs' working to check.

	cannot have negative dimensions.	Lesson 1 table and ask: "Which row matches $w = 10$? What length goes with it?" (12 m , area $10 \times 12 = 120\text{ m}^2$). Then ask: "If $w = -12$, what is the length $w + 2$? Can a hall have a width of -12 metres ?" WAIT TIME of 15 seconds. Cold-call 3 learners for their answer and reason.		
Explain Phase  VIDEO: Zero product property Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: strategy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners construct a written explanation of the hall solution: (1) form the equation $w(w + 2) = 120$; (2) expand to $w^2 + 2w - 120 = 0$; (3) factorise to $(w + 12)(w - 10) = 0$; (4) apply Zero Product Property to get $w = 10$ or $w = -12$; (5) check both in context and reject $w = -12$ because a width cannot be negative; (6) state 'width 10 m , length 12 m , area $10 \times 12 = 120\text{ m}^2$ '. Then, in pairs, they solve the Kisumu fish pond problem independently: width $w\text{ m}$, length $(w + 3)\text{ m}$, area 40 m^2 . They write $w(w + 3) = 40$, $w^2 + 3w - 40 = 0$, $(w + 8)(w - 5) = 0$, $w = 5$ or $w = -8$, reject -8 , and state width 5 m , length 8 m , check $5 \times 8 = 40$. Pairs swap books and peer-check using a checklist: Did they set each factor to zero? Did they check in context? Did they state units?	Say: "You are the committee's advisers. Write a short report for them with the six steps. Use the words 'Zero Product Property' and 'reject' in your report." Display a six-step skeleton. Circulate and ask: "What does the -12 mean in the equation, and what does it mean for the hall?" Cold-call 3 learners to read step 5 aloud and justify it. Say: "Now the Kisumu fish farmer has a pond with an area of 40 m^2 and a length 3 m more than the width. Use the same steps. You have 6 minutes." WAIT TIME of 15 seconds after reading the problem before pairs begin. For pairs who are stuck on factorising $w^2 + 3w - 40$, prompt: "What two numbers multiply to -40 and add to 3 ?" (8 and -5). Do not give the answer if they can find it. Pose: "Both $w = 5$ and $w = -8$ make the equation true. Why do we only use one?" Cold-call 3 pairs. Ask the class: "What is the same in the hall and the pond? What is different?" Have 2 pairs present their work on the board and ask the class to hold up thumbs for agree/disagree.	Structured written explanation and Compare-and-Contrast: Writing a step-by-step report forces learners to link each algebraic step to its meaning in the hall. Comparing the hall with the Kisumu pond (same structure: length = width + k , area given) helps learners see that the method transfers to a different Kenyan context, and the repeated rejection of a negative solution builds the big idea that context decides.	Observable indicators: (1) the written report contains all six steps in order; (2) the pond equation is correctly written as $w^2 + 3w - 40 = 0$; (3) factorisation $(w + 8)(w - 5)$ is correct; (4) both solutions $w = 5$ and $w = -8$ are found and -8 is explicitly rejected with a reason; (5) final answer states 'width 5 m , length 8 m ' with units and a check $5 \times 8 = 40$. Use the peer-check checklist plus teacher sampling of at least 6 pairs. Misconception to watch: learners who give both 5 and -8 as pond dimensions, or who forget to find the length.
Driving Question Board (DQB) Creation  VIDEO: Solving Real-	Learners revisit the class Driving Question Board with the question: 'How can we use quadratic equations to find unknown dimensions and quantities that help our community solve real	Say: "Take 2 minutes. Write one 'What we notice' card and one 'What we wonder' card about today's work. Use the hall or the Kisumu pond in your card." WAIT TIME of 10 seconds before	Driving Question Board with notice/wonder cards (tracking growing understanding): Moving cards from 'wonder' to 'answered' makes learning visible and shows learners that their questions drive	Observable indicators: (1) at least 80% of pairs post a notice card that mentions two solutions or rejecting a solution; (2) wonder cards that are genuine questions about method limits (e.g. non-

World Problems Using Quadratic Equations: A Sample Application Source: CK-12 Search ARES for similar videos EXERCISE: Quadratics by taking square roots: with steps <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	planning problems, like building the Harambee hall in Githurai?' Each pair writes at least one 'What we notice' card (e.g. 'We noticed that $(w + 12)(w - 10) = 0$ gives two answers, 10 and -12') and one 'What we wonder' card (e.g. 'What if the equation does not factorise?' or 'Is there another way to get the same answer?'). They place cards on the board and move earlier question cards that have now been answered (for example, 'How do we find w?') into an 'Answered' column with a note on the evidence.	collecting. Invite 4 pairs to post cards and read them aloud. Ask: "Which earlier question on our board did we answer today? What evidence do we have?" Cold-call 2 learners to move cards. Highlight a 'wonder' that points to Lesson 5, such as, "Can we solve $x^2 + 6x = 48$? It does not seem to factorise." Say: "Keep that question. We will return to it in our next lesson."	the unit. New wonder cards (such as equations that will not factorise) create purpose for the next lesson.	factorising equations); (3) learners cite evidence when moving a card. Record the number of pairs whose cards refer to context-based rejection as a measure of the main idea.
Model Building Phase VIDEO: Strategy in solving quadratic equations <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Use Square Roots to Solve Quadratic Equations (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners revise the cumulative class model poster on manila paper, which already shows the hall sketch, $w(w + 2) = 120$, the general form $a = 1$, $b = 2$, $c = -120$, the expansion and the factorisation. In groups of four they add a new section titled 'Solving by factorisation' with: the Zero Product Property in words, the hall working ($w = 10$ or $w = -12$; reject -12; width 10 m, length 12 m), a small box for the Kisumu fish pond ($w = 5$ m, length 8 m), and the revised model statement: 'A quadratic equation can have two solutions; context tells us which one to use.' They add a note of what the model cannot yet do: solve equations that do not factorise.	Say: "Look at our model poster. What new idea do we need to add today? Talk in your group for 1 minute." WAIT TIME of 10 seconds, then cold-call 3 groups for suggestions. Then say: "Add a section for factorisation. Include the Zero Product Property in your own words, and show both solutions for the hall and which one we keep." Circulate and ask: "Where on your model does the -12 appear, and where is the reason for rejecting it?" Ask each group to write one sentence on 'What our model cannot do yet'. Close by reading the revised statement together: "A quadratic equation can have two solutions; context tells us which one to use." Ask: "Thumbs up, sideways, or down: can I use this idea to help the Githurai committee?"	Cumulative model revision: Adding the new idea to a single poster each lesson shows how knowledge builds from expansion (Lesson 2) to factorisation (Lesson 3) to solving (Lesson 4). Naming the model's limits keeps learners curious and motivates the next method.	Observable indicators: (1) the poster includes the Zero Product Property stated correctly; (2) both $w = 10$ and $w = -12$ appear, with -12 rejected and a reason; (3) the final hall answer is written as 'width 10 m, length 12 m'; (4) the revised statement is present; (5) each group names a limitation. Use a quick exit ticket: 'Solve $x^2 + 4x - 21 = 0$ by factorisation' $((x + 7)(x - 3) = 0, x = -7$ or $x = 3)$ for those who finish early, to check transfer.

D. TEACHER REFLECTION

After the lesson, consider: (1) How many pairs correctly set each factor equal to zero rather than to 120, and what did the errors reveal about their understanding of the Zero Product Property? (2) Did learners justify rejecting $w = -12$ with context reasoning or just because 'the teacher said so'? Which questions drew out the best reasoning? (3) Was the WAIT TIME of 10 to 15 seconds long enough for quieter learners to contribute, and did cold-calling spread participation across the class? (4) Which learners struggled to factorise $w^2 + 3w - 40$ and need more practice with factor pairs from Lesson 3? (5) Which wonder cards (for example about equations that do not factorise) should be used to launch Lesson 5 on completing the square? (6) How could the Kisumu fish pond or another local example be adapted for learners with less experience of fish farming, for instance using a Kericho tea plot or a Gikomba stall?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that when the factorised hall equation $(w + 12)(w - 10) = 0$ is written, one of the brackets must be zero, which gives $w = 10$ or $w = -12$. We saw that $w = 10$ matches the 10 m by 12 m row (area 120 m ²) in the trial-and-error table, and that $w = -12$ would give negative dimensions. In the Kisumu fish pond, $w^2 + 3w - 40 = 0$ gave $w = 5$ or $w = -8$.
What did I learn?	We learned the Zero Product Property: if $A \times B = 0$ then $A = 0$ or $B = 0$. To solve a quadratic equation by factorisation we write it in the form $ax^2 + bx + c = 0$, factorise, set each factor equal to zero, solve for the two solutions, and check each one in the context.
How does this explain the phenomenon?	A quadratic equation can have two solutions, but context tells us which one to use. For the Githurai Harambee hall, $w = 10$ and $w = -12$ both satisfy the equation, but a width cannot be negative, so we reject -12 and conclude the hall has width 10 m and length 12 m (area $10 \times 12 = 120$ m ²). In the same way, the Kisumu fish pond has width 5 m and length 8 m ($5 \times 8 = 40$ m ²).

LESSON 5 (40 minutes (one double lesson may be used if available)): Solving by Completing the Square: The Nakuru School Garden

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Students learn that when a quadratic equation does not factorise over integers, completing the square still solves it. They connect the algebra to an area diagram, so the Githurai committee's tools now work on equations that factorisation cannot handle.
Knowledge	Rewrite $x^2 + bx = k$ as $(x + b/2)^2 = k + (b/2)^2$ and take square roots to get two solutions. Know that $x^2 + 6x + 9 = (x + 3)^2$ is a perfect square trinomial and that the added 9 is $(b/2)^2$. Interpret the steps as completing a square in an area diagram.
Skills	Modelling (NGSS SEP 2): draw and use an area diagram of an x by x square, two 3 by x rectangles and a 3 by 3 corner square. Analysing and interpreting data (SEP 4): test whether $x^2 + 6x - 48 = 0$ factorises and see that it does not. Constructing explanations (SEP 6): justify why the same number is added to both sides. Using mathematics (SEP 5): solve, round to 2 d.p., check in context and reject the negative solution.
Attitudes	Perseverance when the first method (factorisation) fails. Appreciation that mathematics offers more than one method. Collaboration and respect for peers' reasoning in pairs. Responsibility in checking answers against the context.
Key Inquiry Question	How can we use quadratic equations to model and solve problems in day-to-day activities? (KICD) Sub-question: What can we do when a quadratic equation will not factorise?
Purpose in Storyline	Lesson 4 solved by factorisation using the Zero Product Property (hall $w = 10$ or -12 , Kisumu pond $w = 5$). Lesson 5 shows a real garden equation where factorisation fails, and completing the square succeeds. This sets up Lesson 6, where completing the square on $ax^2 + bx + c = 0$ becomes the quadratic formula. DQB update: 'Completing the square always works, even when factorisation fails.'
Safety Notes	No hazardous materials. Scissors for cutting paper pieces: cut away from the body, keep them on desks, and collect all offcuts. Keep the classroom walkways clear while students post cards on the DQB.

B. LESSON OVERVIEW

The Githurai committee found $w = 10$ m by factorising. Now the Nakuru school garden club has a plot with width x m, length $(x + 6)$ m and area 48 m^2 , giving $x^2 + 6x = 48$. Students predict whether it factorises, test factor pairs of -48 that sum to 6, and find none. They cut paper pieces to build the area diagram: an x by x square, two 3 by x rectangles and a missing 3 by 3 corner of area 9. Adding 9 to both sides gives $(x + 3)^2 = 57$, so $x = -3 \pm \sqrt{57}$. Students calculate $\sqrt{57} = 7.55$, accept $x = 4.55$ m, reject -10.55 , find length 10.55 m, and check $4.55 \times 10.55 = 48 \text{ m}^2$ (approximately). They then update the DQB and the cumulative model with this second method, completing the square, and link it back to the hall.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO:	Students recall the Githurai hall $w^2 + 2w - 120 = 0$, which	Say: 'The Nakuru school garden club has 48 square metres to plant	Predict-Test-Conflict (cognitive conflict): students commit to a	Look for: (1) writing $x(x + 6) = 48$ and $x^2 + 6x - 48 = 0$ correctly ($a =$

Solve by completing the square: Integer solutions Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Solve equations by completing the square Source: Khan Academy (English - US curriculum) Search ARES for similar readings	factorised as $(w + 12)(w - 10) = 0$. The teacher shows the Nakuru school garden club's sketch: width x m, length $(x + 6)$ m, area 48 m^2, so $x(x + 6) = 48$, i.e. $x^2 + 6x = 48$. Individually, students write a prediction on a mini-whiteboard or in their notebooks: (1) Will $x^2 + 6x - 48 = 0$ factorise like the hall equation? (2) Roughly what is the width? Then they test in pairs, listing factor pairs of -48 and checking whether any pair has sum 6. Pairs: 1 and -48, 2 and -24, 3 and -16, 4 and -12, 6 and -8 and so on. None sums to 6.	sukuma wiki and onions. The length is 6 m more than the width. Before we calculate anything, write: will this factorise like our hall equation? Yes or no, and why? Give WAIT TIME of 10-15 seconds in silence. Cold-call 3 students for predictions, asking each 'What makes you think so?' Then say: 'Test it. Find two integers with product -48 and sum +6. You have 3 minutes.' Circulate and note pairs who try all factor pairs systematically. Cold-call 2 pairs: 'What did you find?' Record on the board: 'No integer pair works. Factorisation fails here.' Ask: 'The garden still exists. So what now?'	prediction, test it, and experience a method failing. This creates a real need for a new method rather than a teacher-announced one, and links to the hall, where the same approach worked.	1, $b = 6$, $c = -48$); (2) a systematic list of factor pairs of -48 rather than random guessing; (3) pairs stating that no integer pair has product -48 and sum 6. Misconception to watch: students who conclude 'no solution exists' because it does not factorise. Redirect: 'The garden exists, so a width must exist.'
Observe Phase VIDEO: Solving quadratics by completing the square: no solution Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Completing the square (intro) Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Each pair gets a paper kit (cut from old exercise-book or manila paper): one x by x square (use $x = 5$ cm for the paper, labelled x), two 3 by x rectangles (3 cm by 5 cm), and a loose 3 by 3 square of area 9. Step 1: arrange the x by x square and the two 3 by x rectangles in an L-shape. This shape has area $x^2 + 6x$, which equals 48. Step 2: notice the missing corner. Step 3: place the 3 by 3 square into the corner to form a big square of side $(x + 3)$. Step 4: write the area story: $x^2 + 6x + 9 = (x + 3)^2$. Since $x^2 + 6x = 48$, the big square has area $48 + 9 = 57$. Students sketch the diagram in notebooks and label all lengths and areas. Then they take the square root: $x + 3 = \sqrt{57}$ or $x + 3 = -\sqrt{57}$.	Say: 'Your L-shape is the garden's area of 48. It is missing a corner. What area do we need to add to make a perfect square?' WAIT TIME 10-15 seconds. Cold-call 3 students: 'How do you know the corner is 3 by 3?' Press: 'Where does the 3 come from?' (Half of 6.) 'Why is the corner area 9?' Display the board model: $x^2 + 6x = 48$; add 9 to both sides; $x^2 + 6x + 9 = 57$; $(x + 3)^2 = 57$. Ask: 'Why add 9 to BOTH sides?' Cold-call 2 students. Then ask: 'If a number squared equals 57, what could the number be?' WAIT TIME 10 seconds. Elicit the plus and minus square roots: $x + 3 = \sqrt{57}$ or $x + 3 = -\sqrt{57}$. Link to Lesson 3: 'Where have we seen $x^2 + 6x + 9 = (x + 3)^2$ before?' (Perfect square trinomial.)	Concrete-Representational-Abstract (CRA) with the Modelling practice: the paper pieces (concrete) become the sketch (representational), then the algebra (abstract). Seeing the missing 3 by 3 corner explains why we add $(b/2)^2$, so the rule is understood rather than memorised.	Observable indicators: (1) the L-shape is built correctly with two 3 by x rectangles on adjacent sides of the x by x square; (2) students identify the corner as 3 by 3 = 9 and state that 3 is half of 6; (3) students write $(x + 3)^2 = 57$ and give both square roots. Misconception to watch: writing $(x + 3)^2 = x^2 + 9$ (missing the middle term $6x$). Use the diagram to show the two rectangles, which account for $6x$.
Explain Phase VIDEO: Solve by completing the square: Non-integer solutions	Students complete the solution in pairs. From $(x + 3)^2 = 57$: $x = -3 + \sqrt{57}$ or $x = -3 - \sqrt{57}$. With $\sqrt{57} = 7.55$ (2 d.p.), $x = -3 + 7.55 = 4.55$ or $x = -3 - 7.55 = -10.55$. Context decides: a width cannot	Say: 'We have two solutions again, just like the hall. Which one does the Nakuru garden club use, and why?' WAIT TIME 10-15 seconds. Cold-call 3 students. Ask: 'Which solution would the hall committee	Claim-Evidence-Reasoning (CER) plus Compare and Contrast Methods: students state the claim (garden is 4.55 m by 10.55 m), cite the evidence (diagram, $(x + 3)^2 = 57$, check), and give the reasoning	Observable indicators: (1) both roots are written, $-3 + \sqrt{57}$ and $-3 - \sqrt{57}$; (2) rounding to 2 d.p. gives 4.55 and -10.55; (3) -10.55 is rejected with a reason tied to the garden; (4) length 10.55 m is

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Solve equations by completing the square Source: Khan Academy (English - US curriculum) Search ARES for similar readings	be negative, so reject -10.55 and keep $x = 4.55$ m. Length = $x + 6 = 10.55$ m. Check: $4.55 \times 10.55 = 48.0 \text{ m}^2$ (approximately). Then they write a general recipe in their own words for $x^2 + bx = k$: (1) halve b; (2) square it; (3) add to both sides; (4) write $(x + b/2)^2 = k + (b/2)^2$; (5) take square roots (plus and minus); (6) solve for x; (7) check in context and reject any invalid solution. Practice: complete the square for $x^2 + 4x = 5$ (this one also factorises, as $(x + 5)(x - 1) = 0$, giving $x = 1$ or -5) and compare with factorising to confirm both methods agree. Then $x^2 + 8x = 20$ as a second practice. Finally, each pair writes a Claim-Evidence-Reasoning statement: 'The Nakuru garden is 4.55 m wide and 10.55 m long because...'	reject, and why is it the same reason?' (Negative width.) Say: 'Check by multiplying 4.55 by 10.55. Do you get exactly 48?' Elicit that $4.55 \times 10.55 = 48.0$ approximately because of rounding. Introduce the recipe, with students filling the numbers. For $x^2 + 4x = 5$, ask: 'You solved this by factorising too. Do both methods give the same answer?' Cold-call 2 students. Ask: 'Which method is faster for the Nakuru garden? Which one works for every equation?' Allow 10 seconds WAIT TIME. Circulate and look for sign errors with the negative root.	(context rejects a negative width). Comparing completing the square with factorisation on equations that both can solve builds trust in the new method.	stated and the area check is shown; (5) in $x^2 + 8x = 20$, students add 16 to both sides, giving $(x + 4)^2 = 36$, $x = 2$ or -10. Misconceptions: adding $(b/2)^2$ to only one side; forgetting the negative root; forgetting to subtract 3 after taking the square root. Use quick checks and ask pairs to explain errors to each other.
Driving Question Board (DQB) Creation VIDEO: Solve by completing the square: Integer solutions Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Completing the square (intro) Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Students write 'What we notice / What we wonder' cards on the DQB. Examples of what they might notice: 'Factorisation failed for the Nakuru garden but completing the square worked'; 'The 9 we add is half of 6 squared'; 'The answer has a square root in it'. Examples of what they might wonder: 'Can we complete the square for any quadratic, even when the x^2 has a number in front?'; 'Is there a faster method that always works?'; 'How do we know which method to pick?' The class agrees on a DQB update: 'Completing the square always works, even when factorisation fails.' They also move answered cards (e.g. 'What if the equation does not factorise?') to the answered column.	Say: 'Take 2 minutes to write one notice card and one wonder card about today.' WAIT TIME 10-15 seconds before pairs start sharing. Cold-call 4 students to read cards aloud and post them on the board. Ask: 'Which earlier wonder cards did today's lesson answer?' Cold-call 2 students to move cards. Say: 'Read the new statement: Completing the square always works, even when factorisation fails. Do we agree? What evidence do we have? What do we still need to test?' Highlight that the claim is supported by one example only so far, and mention that Lesson 6 will test whether the steps can be made into a formula for any equation.	Driving Question Board with notice/wonder cards: students track their own questions and evidence, which makes learning visible and shows the next steps. Naming the limits of current evidence (one example) models scientific and mathematical caution.	Look for: (1) cards that refer to specific evidence (the 3 by 3 corner, $(x + 3)^2 = 57$, 4.55 m); (2) wonder cards pointing to the next idea (a general method, a leading coefficient not equal to 1); (3) students who justify moving a card using evidence rather than opinion. Note the students who write only 'it was fun' and ask them to add a mathematical detail.
Model Building	Students update the cumulative model poster for the Githurai hall	Say: 'Our model must now include a second method. Add the Nakuru	Cumulative Model Building with Cross-Validation: students extend	Observable indicators: (1) the poster shows the diagram and

Phase  VIDEO: Solving quadratics by completing the square: no solution Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: strategy <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	problem. New section: 'Method 2: Completing the square.' Each group adds: (1) the Nakuru diagram with the 3 by 3 corner; (2) the steps $x^2 + 6x = 48$, $x^2 + 6x + 9 = 57$, $(x + 3)^2 = 57$, $x = -3 \pm \sqrt{57}$; (3) the decision: 4.55 m, reject -10.55; (4) a 'when to use it' note: it works when factorisation fails. Then groups apply completing the square to the hall: $w^2 + 2w = 120$. Halve 2 to get 1, add 1: $w^2 + 2w + 1 = 121$, $(w + 1)^2 = 121$, $w + 1 = 11$ or -11, so $w = 10$ or $w = -12$. They compare with the factorisation result from Lesson 4 and confirm that both give width 10 m, length 12 m. Individual exit ticket: solve $x^2 + 10x = 11$ by completing the square (add 25; $(x + 5)^2 = 36$; $x = 1$ or -11).	diagram and the steps. Then test your model on the hall: do you get the same width as factorisation?' Give WAIT TIME of 10-15 seconds before groups begin the hall check. Cold-call 3 groups to report their result for w. Ask: 'Why does it matter that two different methods give the same answer?' Cold-call 2 students. Say: 'Add a rule to your poster: always check in context.' For the exit ticket, collect responses and sort them into correct, sign error, and did-not-add-to-both-sides piles to plan Lesson 6.	the class model with a new method and test it on the original phenomenon. Getting the same hall answer (width 10 m, length 12 m) with two methods builds confidence that the model is consistent.	algebra aligned ($x^2 + 6x + 9 = (x + 3)^2$); (2) the hall check gives $(w + 1)^2 = 121$ and $w = 10$, with -12 rejected; (3) the exit ticket gives $x = 1$ or -11 with correct completion (add 25). Success criterion: at least 80% of students complete the square correctly with both solutions. Re-teach with the diagram for students who add the wrong number.
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D. TEACHER REFLECTION

Which students used the paper diagram to explain why 9 is added, and which only followed the steps? Did anyone try to complete the square for an equation where the coefficient of x is odd, and how did the class respond? How many pairs forgot the negative square root or forgot to subtract 3, and what did the diagram or hall comparison do to correct this? Did the failure of factorisation on the Nakuru garden create genuine curiosity, or did students see it as a trick? What would you change for Lesson 6, where the same steps are applied to the general equation and the quadratic formula emerges?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that $x^2 + 6x - 48 = 0$ (the Nakuru school garden, area 48 m ² , length 6 m more than the width) has no integer factor pair with product -48 and sum 6, so it does not factorise. We also observed with paper pieces and a diagram that an x by x square and two 3 by x rectangles need a 3 by 3 corner (area 9) to form a square of side (x + 3).
What did I learn?	We learned that $x^2 + bx = k$ can be rewritten as $(x + b/2)^2 = k + (b/2)^2$ and solved by taking square roots, giving two solutions. For the garden, $(x + 3)^2 = 57$, so $x = -3 \pm \sqrt{57}$, which is 4.55 or -10.55. Context tells us to reject the negative width.
How does this explain the phenomenon?	The garden is 4.55 m wide and 10.55 m long, because a width cannot be negative and $4.55 \times 10.55 = 48$ m ² (approximately). Completing the square works when factorisation fails, and it also gives the hall answer: $(w + 1)^2 = 121$, so $w = 10$ or -12, and the Githurai hall is width 10 m, length 12 m.

LESSON 6 (80 minutes (2 x 40-minute periods)): The Quadratic Formula and the Mombasa Stone Problem

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners derive the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ by completing the square on $ax^2 + bx + c = 0$, use it to solve equations that arise in real planning problems, and use the discriminant $D = b^2 - 4ac$ to decide how many real solutions an equation has.
Knowledge	Learners will know that completing the square on the general form $ax^2 + bx + c = 0$ gives $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, that the discriminant $D = b^2 - 4ac$ tells us the number of real solutions ($D > 0$ two real solutions, $D = 0$ one repeated solution, $D < 0$ no real solutions), and that context decides which solution is valid.
Skills	Learners will: (1) derive the quadratic formula step by step (SEP: Constructing Explanations and Developing and Using Models); (2) identify a, b, c and substitute carefully, including negative values; (3) use calculators to compute and verify solutions (SEP: Using Mathematics and Computational Thinking); (4) compare results from factorisation, completing the square and the formula (SEP: Analysing and Interpreting Data); (5) reject impossible solutions using context (SEP: Engaging in Argument from Evidence).
Attitudes	Learners will appreciate that one general tool can solve every quadratic equation (Unity and Responsibility in group work and when caring for shared calculators), show perseverance when working with negative numbers and square roots, and value quadratic thinking in community planning (Life Skills: critical thinking and self-esteem through explaining their reasoning to peers). Core competencies: Critical Thinking and Problem Solving, Communication and Collaboration, Digital Literacy (calculator use).
Key Inquiry Question	1. How do quadratic expressions and equations differ from linear ones, and why are they necessary? 2. How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	Lesson 5 showed that completing the square solves equations that do not factorise, such as the Nakuru garden $x^2 + 6x = 48$. That method takes many steps each time. In this lesson learners apply completing the square once to the general form $ax^2 + bx + c = 0$ and obtain a universal formula. They then test it on the Mombasa stone problem and Mama Wanjiru's mandazi pricing, and check that it reproduces the Githurai hall answer ($w = 10$ m). This advances the driving question by giving the Harambee committee a method that works for any quadratic equation, and sets up Lesson 7, where learners choose and justify the most efficient method.
Safety Notes	Mombasa stone problem is modelled on paper only; no one throws objects in class. Learners handle shared calculators carefully and return them at the end. Group seating should leave clear walkways. Be sensitive to learners who struggle with negative numbers; use pair support and avoid public criticism of errors.

B. LESSON OVERVIEW

The lesson opens with a prediction: the committee in Githurai wants one tool that works every time, so learners predict whether a single formula could solve any quadratic equation, and how many solutions the Mombasa stone equation $-5t^2 + 20t + 25 = 0$ will give. In the Observe phase learners work in groups to complete the square on $ax^2 + bx + c = 0$ and so derive $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, then use it on the Mombasa stone ($D = 900$, $t = 5$ s after rejecting $t = -1$) and compare with factorisation. They then check the Githurai hall with $D = 484$ ($w = 10$ or -12), solve Mama Wanjiru's mandazi pricing problem ($D = 200$, $p = 7.93$ or 22.07 KES), and investigate $D = 0$ and $D = -16$. In the Explain phase learners decide which mandazi price to choose and why, and explain what D tells us. The DQB and the cumulative model are updated: the quadratic formula is a universal tool, and it works for any quadratic equation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Number of solutions of quadratic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the committee's manila-paper sketch (width = w , length = $w + 2$, Area = 120 m^2) and recall that $w^2 + 2w - 120 = 0$ was solved by factorisation and that the Nakuru garden needed completing the square. They are then told: a stone is thrown upward from a viewing platform 25 m above the sea in Mombasa, with $h = -5t^2 + 20t + 25$. Working individually for 2 minutes, then in pairs, they write on a half-sheet: (1) Do you think there is one tool that works for every quadratic equation? Why or why not? (2) When the stone hits the sea, $h = 0$. How many answers for t do you expect, and could factorising be hard here? They write a rough guess for the landing time in seconds.	Display the sketch on manila paper and the stone equation. Say: 'The Githurai committee is stuck because w appears multiplied by itself. Last lesson we completed the square for the Nakuru garden. It worked, but it took many steps. Today's question: can we get one tool that works every time?' Ask learners to write predictions silently for 2 minutes. Pose: 'What is a , what is b , and what is c in $-5t^2 + 20t + 25 = 0$?' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 4 learners for their predicted landing time and 3 for their reasons. Record all predictions on the board without judging them. Say: 'Keep your prediction card. We will test it with evidence.'	Predict-Observe-Explain (prediction before evidence): committing to a written guess reveals current ideas (for example that every quadratic needs a different trick, or that a negative time cannot appear). Learners later confront their own predictions with calculated evidence, which makes the conceptual change visible and memorable.	Observe whether learners correctly name $a = -5$, $b = 20$, $c = 25$ (watch for sign errors on a). Note how many predicted two answers versus one. Look for guesses reached by trial and error. Tally on the board: how many predict 'one tool works' versus 'different tool each time'. Use these tallies to target the Observe phase.
Observe Phase  VIDEO: Using the quadratic formula: number of solutions - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE: Quadratic formula Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Part A (derivation, about 20 minutes): In groups of 4, learners get a strip with the steps of completing the square on $ax^2 + bx + c = 0$ in scrambled order. They first recall the Nakuru pattern $x^2 + 6x + 9 = 57$ and then follow the same moves in letters: divide by a to get $x^2 + (b/a)x + c/a = 0$; move c/a across; add $(b/2a)^2$ to both sides; write the left side as $(x + b/2a)^2$; combine the right side to $(b^2 - 4ac)/4a^2$; take square roots with $\pm$; solve for x . They reorder the steps and glue them into their notebooks, finishing with $x = (-b \pm \sqrt{(b^2 - 4ac)})/(2a)$. Part B (Mombasa stone, about 15 minutes): Using $a = -5$, $b = 20$, $c = 25$ they compute $D = 20^2 - 4(-5)(25) = 400 + 500 = 900$, $\sqrt{900} = 30$, and $t = (-20 \pm$	Distribute pre-cut derivation strips (one set per group) and a shared calculator per group. Circulate during the derivation and ask: 'Which line matches the step where we added 9 in the Nakuru garden? What is the letter version of 9?' Give WAIT TIME of 10-15 seconds after each question. Prompt common errors: 'Where did the $4a^2$ come from?' and 'Why do we write $\pm$?' Pause after the derivation: 'Everyone, point to the line where $b^2 - 4ac$ first appears.' Cold-call 3 groups to read out the final formula. For the stone, say: 'Careful with $a = -5$. What is $4 \times (-5) \times 25$? Check the sign before you press equals.' Ask 2 groups to show how they handled $-4ac$. After $t = -1$ and $t = 5$, ask: 'Which	Generalising from a specific case to the general case (worked-example-to-formula, with scrambled-strip sequencing): learners see that the formula is not magic but the same completing-the-square steps done once with letters, which supports conceptual understanding. Data table of D values: comparing several equations helps learners notice the pattern between the sign of D and the number of real solutions (SEP: Analysing and Interpreting Data).	Check that each group's strips are in a valid order and that they can explain the step that adds $(b/2a)^2$. Observable indicators: correct substitution of $a = -5$ giving $D = 900$; correct handling of division by -10 ; rejecting $t = -1$ with a reason about time; the calculator check giving $h = 0$ at $t = 5$. Verify the D table is complete with $D = 900$, 484 , 0 and -16 . Note learners who write ' $D = -16$ so $x = -4$ ' (misconception) and address them directly.

	30)/(-10), giving $t = -1$ or $t = 5$. They compare with factorisation: dividing by -5 gives $t^2 - 4t - 5 = 0 = (t - 5)(t + 1)$. They reject $t = -1$ (before the throw), so the stone lands at $t = 5$ s, and use a calculator to confirm $h = -5(25) + 20(5) + 25 = 0$. They also find the maximum height at $t = 2$ s: $h = -20 + 40 + 25 = 45$ m. Part C (hall check, about 8 minutes): For $w^2 + 2w - 120 = 0$ ($a = 1$, $b = 2$, $c = -120$) they compute $D = 4 + 480 = 484$, $\sqrt{484} = 22$, $w = (-2 \pm 22)/2 = 10$ or -12, matching the factorised answer: width 10 m, length 12 m. Part D (discriminant, about 12 minutes): Groups evaluate D for $x^2 - 6x + 9 = 0$ ($D = 0$, one repeated solution $x = 3$), $x^2 + 2x + 5 = 0$ ($D = -16$, no real solutions, because $\sqrt{-16}$ is not a real number), and record them with the stone ($D = 900$) and hall ($D = 484$) in a table of equation, a, b, c, D, number of real solutions.	answer do you keep, and why?' (recalling rejection from Lesson 4). For $D = -16$ ask: 'What happens when you try $\sqrt{-16}$ on the calculator?' Let learners see the Math Error message. Ask 3 learners what the sign of D tells us.		
Explain Phase  VIDEO: Proof of the quadratic formula Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Number of solutions of quadratic equations Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners apply the formula to Mama Wanjiru's mandazi pricing. She sells $n = 120 - 4p$ mandazi a day at price p KES each and wants a daily revenue of 700 KES. In pairs they write revenue $p(120 - 4p) = 700$, expand to $4p^2 - 120p + 700 = 0$, divide by 4 to get $p^2 - 30p + 175 = 0$, and note $a = 1$, $b = -30$, $c = 175$. They compute $D = 900 - 700 = 200$, $\sqrt{200} = 14.14$, and $p = (30 \pm 14.14)/2 = 7.93$ or 22.07 KES. They check both give revenue 700: at $p = 7.93$ she sells about 88 mandazi, and at $p = 22.07$ about 32. Learners discuss in their pairs which price Mama Wanjiru should pick, and why both are mathematically valid but context helps decide: the lower price sells more mandazi, so she	Introduce: 'Mama Wanjiru in Gikomba-style market trading wants exactly 700 KES a day. Is there only one price?' Let pairs work for 6 minutes. Walk around asking: 'Why did we divide by 4 first?' and 'What is b here? Is it 30 or -30?' Give WAIT TIME of 10-15 seconds. Ask 3 pairs to share their two prices and 2 pairs to justify the choice of 7.93 KES. Challenge: 'Both prices give 700 KES. Why would she not choose 22.07?' Push for evidence using mandazi numbers sold (88 versus 32). Then ask: 'In which situation did we reject a solution, and in which did we keep both?' to contrast the stone (reject $t = -1$) and mandazi (both valid, choose by context). For the Nakuru cross-check say:	Claim-Evidence-Reasoning (SEP: Engaging in Argument from Evidence): learners state a claim about which price to choose, support it with calculated numbers of mandazi sold, and reason from context. Comparing methods (formula versus completing the square) shows that different methods give the same answer, building trust in the formula as a universal tool.	Look for correct values $a = 1$, $b = -30$, $c = 175$, $D = 200$, and prices 7.93 and 22.07 to 2 d.p. Indicators of understanding: learners reference both revenue checks and number sold; they explain that $D > 0$ gives two real solutions; they note the Nakuru result matches. Collect the 3-sentence explanations as an exit ticket; mark with a 3-point checklist (formula generality, role of D, context decides). Identify learners who still reject a valid solution automatically.

	picks 7.93 KES. They then cross-check the Nakuru equation $x^2 + 6x - 48 = 0$: $D = 36 + 192 = 228$, $x = (-6 \pm \sqrt{228})/2 = -3 \pm \sqrt{57}$, which matches the Lesson 5 result of 4.55 m. Finally each learner writes a 3-sentence explanation: (1) why the formula works for any quadratic equation, (2) what D tells us, and (3) how context decides which solution to use. This links back to the hall: the committee now has a tool that works whether or not the equation factorises.	'Compare this with the Lesson 5 answer. Do the two methods agree?' Cold-call 3 learners to read their written explanation aloud. Use a sentence starter on the board: 'The formula works for any quadratic because ...'.		
Driving Question Board (DQB) Creation  VIDEO: Using the quadratic formula: number of solutions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratic formula Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the class DQB, which uses 'What we notice / What we wonder' cards. Individually, each writes one 'What we notice' card (for example: 'D = 900 and the stone lands at 5 s' or 'D = -16 means no real solution') and one 'What we wonder' card (for example: 'Which method is quickest for the Harambee committee?' or 'Why does D = 0 give only one answer?'). They post cards under the driving question about the Harambee hall in Githurai. As a class they read the update line and agree on it: 'The quadratic formula is a universal tool — it works for any quadratic equation.' They also place earlier questions that are now answered (for example 'What do we do when it will not factorise?') with a tick, and group the new cards into themes: using the formula, the discriminant, choosing between solutions.	Say: 'Take two minutes to write one thing you noticed and one thing you wonder.' Give WAIT TIME of 10-15 seconds before learners begin writing. Cold-call 4 learners to read their cards, then ask the class: 'Does this card answer an earlier question on our board?' Move answered questions to a 'Now we know' column. Highlight the new wonder cards that point to Lesson 7: 'Which method is best for which equation?' Write the DQB update on a strip: 'The quadratic formula is a universal tool — it works for any quadratic equation.' Ask 3 learners to explain in their own words what 'universal' means using the stone or mandazi example. Keep it brisk, about 8 minutes.	Driving Question Board (notice/wonder cards): publicly tracking questions makes growing understanding visible, shows learners their own questions drive the unit, and prepares the next investigation, which is choosing efficient methods in Lesson 7.	Check that notice cards contain evidence (numbers such as D = 900 or t = 5 s) rather than vague statements. Check that wonder cards are about mathematics and not just procedure. Count how many learners connect D to the number of solutions. Note whether any cards show the misconception that the formula replaces the need for checking solutions in context.
Model Building Phase  VIDEO: Using the quadratic	Learners add to the class cumulative model poster for the Githurai hall. In groups they extend the poster with a new 'Method 3: quadratic formula' panel showing the hall check: a = 1, b = 2, c = -	Say: 'Our model must now show three ways of finding the hall dimensions: factorisation, completing the square, and the quadratic formula. Add the third panel.' Circulate and ask: 'Where	Cumulative model revision (SEP: Developing and Using Models): adding the formula panel and a decision flow chart lets learners organise three methods into one coherent model of how to solve	Check posters for the correct hall panel (a = 1, b = 2, c = -120, D = 484, w = 10 or -12, reject -12, width 10 m, length 12 m), the flow chart showing all three cases of D, and the context check step.

formula: number of solutions - dubbed(KY) Source: Khan Academy (Kiswahili) Search ARES for similar videos EXERCISE: Solve quadratic equations: complex solutions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	120, $D = 484$, $w = (-2 \pm 22)/2 = 10$ or -12 , rejecting -12 because a width cannot be negative, giving a hall of width 10 m, length 12 m, area 120 m ² . They add a small flow chart: write in general form $ax^2 + bx + c = 0$; find a, b, c; compute D; if $D > 0$ two real solutions, $D = 0$ one repeated solution, $D < 0$ no real solutions; substitute into the formula; check each solution in context and reject impossible ones. They include the Mombasa stone ($D = 900$, lands at $t = 5$ s) and Mama Wanjiru's mandazi ($D = 200$, $p = 7.93$ or 22.07 KES) as side examples. Each learner then completes a short summary table in their notebook (see the prompt) and revises their original prediction card from the start of the lesson, writing what they would now say.	on your poster does the stone reject an answer? Where does the hall reject an answer?' Give WAIT TIME of 10-15 seconds before taking responses. Ask 3 groups to present their flow chart in 1 minute each, and cold-call 2 learners to name what D tells us. Push on precision: 'Does $D = 484$ being a perfect square tell us anything about whether the equation factorises?' (Encourage them to notice that $\sqrt{484} = 22$ is whole, and the equation did factorise.) Remind learners that the model must connect back to the committee's problem and that Lesson 7 will ask them to choose the most efficient method. Collect notebook tables to review.	quadratic equations, and shows that every method leads to the same hall: width 10 m, length 12 m. Revising the original prediction reinforces conceptual change.	Observe whether learners can explain why the three methods agree. Review notebooks: revised prediction should show that the formula works for any quadratic equation, and the table should be filled with accurate data. Flag learners who treat $D < 0$ as a solution.
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D. TEACHER REFLECTION

After the lesson ask yourself: Did learners understand each line of the derivation, or did they only copy the final formula? Which sign errors (for example with $a = -5$ or $b = -30$) were most common, and what pair support helped? Could learners explain why $D = -16$ gives no real solutions, and why $D = 484$ and $D = 900$ give two? Did they reject $t = -1$ and choose $p = 7.93$ KES using context rather than because the teacher told them? Did the DQB update 'The quadratic formula is a universal tool — it works for any quadratic equation' emerge from learners' own evidence? Which learners need more practice before Lesson 7, where they will choose the most efficient method?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	What did we observe? Record the values of a, b, c and D for the Mombasa stone ($-5t^2 + 20t + 25 = 0$), the Githurai hall ($w^2 + 2w - 120 = 0$), Mama Wanjiru's mandazi ($p^2 - 30p + 175 = 0$), $x^2 - 6x + 9 = 0$ and $x^2 + 2x + 5 = 0$, and the number of real solutions each gave.
What did I learn?	What did we learn? Write the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ and explain how it comes from completing the square on $ax^2 + bx + c = 0$. State what $D > 0$, $D = 0$ and $D < 0$ mean for the number of real solutions.
How does this explain the phenomenon?	How does this help explain the phenomenon? Explain how the formula gives the Githurai hall as width 10 m and length 12 m ($D = 484$, $w = 10$ or -12 , reject -12), and how context decides which solution to use in the stone problem ($t = 5$ s) and Mama Wanjiru's pricing (7.93 KES chosen over 22.07 KES).

LESSON 7 (40 minutes (one double lesson of 80 minutes recommended if the poster is completed in class)): Application, Model Building and Consolidation: Closing the Githurai Harambee Hall Story

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate the whole unit: learners choose and justify the most efficient method (factorisation, completing the square or quadratic formula) in new Kenyan contexts, write the Final Explanation for the Githurai Harambee hall, compare their Lesson 1 prediction with their final model, and formally close the Driving Question Board. No new technique is taught.
Knowledge	By the end of the lesson, the learner should be able to: (1) solve the Githurai hall equation $w(w + 2) = 120$, i.e. $w^2 + 2w - 120 = 0$, by all three methods and show that each gives $w = 10$ or $w = -12$; (2) explain that context selects the valid solution, so the hall is width 10 m, length 12 m; (3) apply quadratic equations to Kitale land ($2w^2 + w - 78 = 0$), Harambee ticket revenue ($p^2 - 30p + 200 = 0$) and the hall verandah ($x^2 + 11x - 12 = 0$); (4) explain how quadratic expressions and equations differ from linear ones and why they are necessary.
Skills	NGSS-aligned practices adapted for mathematics: Constructing Explanations and Designing Solutions (writing a Final Explanation with claim, evidence, reasoning); Developing and Using Models (building and comparing the Lesson 1 model with the final cumulative model); Engaging in Argument from Evidence (justifying a preferred method); Using Mathematics and Computational Thinking (selecting methods, checking in context); Communication and Collaboration (group presentation); Critical Thinking and Problem Solving (choosing and justifying methods).
Attitudes	Appreciate the use of quadratic equations in solving community planning problems (Harambee, land use, fundraising); Unity: working together as a committee and as groups; Responsibility: checking each solution against the context before giving a final answer; Life Skills (self-esteem): confidently presenting and defending a method choice.
Key Inquiry Question	1. How do quadratic expressions and equations differ from linear ones, and why are they necessary? 2. How can we use quadratic equations to model and solve problems in day-to-day activities?
Purpose in Storyline	Lesson 1 left the Githurai committee stuck with $w(w + 2) = 120$ because arithmetic and linear equations could not find w . Lessons 2 to 6 gave learners expansion, factorisation, the Zero Product Property, completing the square and the quadratic formula with the discriminant. In Lesson 7 learners return to the full hall problem, solve it three ways, apply their thinking to new problems, build the final cumulative model and close the DQB, so they can answer the Driving Question: how quadratic equations help the community find unknown dimensions and quantities.
Safety Notes	Low-risk lesson. Keep walkways clear during gallery walk. Share calculators and manila paper/markers responsibly (Responsibility). Handle scissors or tape for poster work with care. Ensure every learner, including those who struggle, has a role in the group so no one is excluded.

B. LESSON OVERVIEW

Learners begin by revisiting their own Lesson 1 prediction for the width of the Githurai Harambee hall (area 120 m², length = $w + 2$) and the manila-paper sketch. They then solve $w^2 + 2w - 120 = 0$ by factorisation, $(w + 12)(w - 10) = 0$; by completing the square; and by the quadratic formula ($D = 484$, $\sqrt{22}$), and confirm $w = 10$ or $w = -12$, rejecting -12 because a width cannot be negative, so the hall is width 10 m, length 12 m. In groups they solve new problems: Kitale land ($2w^2 + w - 78 = 0$, $w = 6$, length 13 m), Harambee ticket fundraising ($p^2 - 30p + 200 = 0$, $p = 10$ or 20 KES, choose 10 KES), and the hall verandah ($x^2 + 11x - 12 = 0$, $x = 1$ m), and justify their method choice. They write the Final Explanation, build a cumulative model poster or digital slide comparing the Lesson 1 model with the final model, and formally close the Driving Question Board by matching every 'What we wonder' card to an answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  HTML: Applications of Quadratic Equations (Flexbook) Source: CK-12  Search ARES for similar readings	Learners look again at the Githurai committee's manila-paper sketch (width = w , length = $w + 2$, Area = 120 m ²) and at their own Lesson 1 width predictions. Individually they write a quick prediction: 'For the three new problems (Kitale land, ticket price, hall verandah), which method would be most efficient and why?' They also predict: 'Will each problem have one answer or two, and will both be usable?' Then they privately rate their confidence (1-3) on solving the hall equation by each of the three methods.	Display the manila-paper sketch and the Lesson 1 trial-and-error table ($w = 8$, area 80; $w = 9$, 99; $w = 10$, 120; $w = 11$, 143; $w = 12$, 168). Say: 'In Lesson 1 the Githurai committee was stuck. Today we close their story. Look at your first prediction of the width. Without calculating, how do you feel about it now?' Give WAIT TIME of 10-15 seconds. Then pose: 'Here are three new problems. Before we solve anything, predict which method you would choose for each and write one reason.' Cold-call 3 learners to share predictions without judging them; record the predictions on the board under 'Method Predictions'. Circulate and note learners who choose trial and error by habit.	Predict-Reflect-Compare: learners revisit a prior prediction and make a new one about method choice. Seeing how their thinking changed from 'guess and check' to 'choose a method' makes their growth visible and creates purpose for the application work.	Observable indicators: (1) learners can state their Lesson 1 prediction and say whether it was close to $w = 10$; (2) at least 3 of 5 cold-called learners give a reason linked to the equation's form (for example, 'it factorises easily' or 'the numbers are awkward'); (3) learners record a prediction about one vs two solutions. Note learners who still refer only to trial and error and prompt them: 'What does the w times w tell you?'
Observe Phase  VIDEO: Solving Real-World Problems Using Quadratic Equations: A Sample Application Source: CK-12  Search ARES for similar videos  HTML: Applications of Quadratic Equations (Flexbook) Source: CK-12  Search ARES for similar	Working in groups of 4 as the 'Githurai Harambee committee', learners gather evidence by solving the full hall problem three ways on a shared manila sheet divided into three columns. (1) Factorisation: $w^2 + 2w - 120 = 0$ becomes $(w + 12)(w - 10) = 0$, so $w = -12$ or $w = 10$ using the Zero Product Property. (2) Completing the square: $w^2 + 2w = 120$, add 1 to get $(w + 1)^2 = 121$, $w + 1 = \pm 11$, $w = 10$ or -12 . (3) Quadratic formula with $a = 1$, $b = 2$, $c = -120$: $D = 4 + 480 = 484$, $\text{sqrt} = 22$, $w = (-2 \pm 22)/2 = 10$ or -12 . They record steps, count steps and time taken for each method, and check $w = 10$ in the context: length 12 m,	Say: 'You are the Githurai committee. Show them three ways to find the width, and compare how much work each takes.' Assign roles: reader, recorder, checker, reporter. Circulate with targeted prompts: 'What are a , b and c in your equation?', 'Which factor pair of -120 adds to 2?', 'Show me where you rejected a solution and why.' For Kitale land, ask: 'Why is $a = 2$ here, and what does that do to your factor search?' For tickets, ask: 'If both $p = 10$ and $p = 20$ give KES 400, which would the committee choose?' For the verandah: 'Which solution is a length you could actually build?' Give WAIT TIME of 10-15 seconds	Comparing Multiple Representations/Methods (Three-Column Method Comparison): placing factorisation, completing the square and the quadratic formula side by side for the same equation shows that different methods give identical solutions and lets learners judge efficiency, building understanding that the method is a tool while the context decides the valid answer.	Observable indicators: (1) every group's three columns end with $w = 10$ or $w = -12$; (2) the checker verifies $10 \times 12 = 120$ m ² ; (3) groups correctly state a , b , c for each new equation (Kitale $a = 2$, $b = 1$, $c = -78$; tickets $a = 1$, $b = -30$, $c = 200$; verandah $a = 1$, $b = 11$, $c = -12$); (4) groups reject -6.5 m and -12 m with a context reason; (5) groups state that both ticket prices work mathematically. Use a quick tally of correct answers per station; re-teach in a mini-clinic for groups that mis-expanded $(12 + 2x)(10 + 2x)$.

readings	10 x 12 = 120 m ² . They then use the three new problems as additional evidence in a rotating 'station' format: (a) Kitale land, width w m, length (2w + 1) m, area 78 m ² ; (b) Harambee ticket price p KES with tickets sold 60 - 2p and revenue 400 KES; (c) hall verandah with uniform border x m around the 12 m by 10 m hall, outer area 168 m ² .	after each question before offering hints. Cold-call 4 groups to report one method result each. Provide the answer key only after each group has checked in context: Kitale $(2w + 13)(w - 6) = 0$, w = 6 (reject -6.5), length 13 m, 6 x 13 = 78; Tickets $p^2 - 30p + 200 = 0$, (p - 10)(p - 20) = 0, p = 10 gives 40 tickets, p = 20 gives 20 tickets; Verandah $(12 + 2x)(10 + 2x) = 168$ leads to $x^2 + 11x - 12 = 0$, (x + 12)(x - 1) = 0, x = 1 m (reject -12), outer 14 m by 12 m = 168 m ² .		
Explain Phase  VIDEO: Solving quadratic equations: complex roots <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve quadratic equations: complex solutions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group prepares a 2-minute justification: 'Which method would you choose for each problem, and why?' They compare efficiency: factorisation is quickest when integers work (hall, Kitale, tickets, verandah); completing the square helps when the equation does not factorise nicely (like the Nakuru garden); the quadratic formula always works and the discriminant (D = 484 for the hall) shows the number of real solutions. Then each learner writes the Final Explanation individually using Claim-Evidence-Reasoning: Claim: the hall is width 10 m and length 12 m. Evidence: trial-and-error table, $(w + 12)(w - 10) = 0$, D = 484, 10 x 12 = 120 m ² . Reasoning: w ² appears because the unknown multiplies itself, so the equation is quadratic and has two solutions, but a width cannot be negative, so context selects w = 10. They also answer: 'How is this different from a linear equation, such as the perimeter 4w + 4?'	Say: 'Convince the Githurai committee. Which method would you recommend and why? Use evidence, not just opinion.' Give WAIT TIME of 10-15 seconds before groups talk. Cold-call 3 groups to present; after each, ask a different group: 'Do you agree or disagree, and what evidence supports that?' Press for reasoning: 'Why does the quadratic formula always work but not always get used first?' Show a sentence frame on the board: 'I claim ____ because ____, and the evidence is ____, which shows ____.' Address misconceptions: some learners think both ticket prices must be wrong because there are two answers; clarify that both give KES 400 but the committee chooses p = 10 because 40 people attend instead of 20. Contrast linear and quadratic: perimeter 4w + 4 is linear, highest power 1, one solution; area w(w + 2) has highest power 2, up to two solutions. Collect Final Explanations for review.	Claim-Evidence-Reasoning (CER) with Peer Argumentation: learners must connect the claim (10 m by 12 m) to evidence from several methods and explain with the idea that context selects the valid solution. Defending choices to peers builds deeper understanding of why quadratic equations are necessary.	Observable indicators using a 4-point checklist for the Final Explanation: (1) correct equation $w^2 + 2w - 120 = 0$; (2) at least one correct solution method shown with both solutions w = 10 and w = -12; (3) explicit rejection of -12 because a width cannot be negative; (4) a statement contrasting quadratic with linear (highest power 2 vs 1). Learners scoring 3-4 are secure; 0-2 receive a conference during the model-building phase. Listen for the phrase 'context decides which solution to use'.
Driving Question	Learners return to the Driving Question Board with the 'What we	Say: 'Our board started with questions about the Githurai hall.	Driving Question Board Closure (Question-to-Evidence Mapping):	Observable indicators: (1) all original wonder cards have an

Board (DQB) Creation  VIDEO: Proof of the quadratic formula <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve equations by completing the square <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	notice / 'What we wonder' cards from earlier lessons (for example, 'Why can't we just use a linear equation?', 'Why are there two answers?', 'How do we find the width exactly?', 'Can every quadratic be factorised?'). In groups they take turns choosing a card, reading it aloud, and attaching an answer card with evidence from the unit (trial-and-error table, factorisation, completing the square, formula, discriminant, Kitale, tickets, verandah). They add 'What we now understand' cards and one new wonder card for future learning (for example, the maximum height of the Mombasa stone or other curves). The DQB is formally closed when every original wonder card has an answer.	Let us answer each one with evidence.' Hold up each unanswered card and ask: 'Which lesson or problem helps us answer this?' Give WAIT TIME of 10-15 seconds. Cold-call 5 learners, one per card, to read the question and answer. Push on weak answers: 'What evidence from the hall problem supports that?' If a card is unanswered, ask: 'What would we need to find out?' and move it to a 'Future Questions' section. Finally, read the Driving Question aloud and ask 3 learners to summarise an answer in one sentence each. Praise correct use of terms: solutions, roots, discriminant.	linking each original wonder card to specific evidence shows learners how their questions guided the unit and consolidates the idea that quadratic equations model situations in which an unknown is multiplied by itself.	answer card with named evidence; (2) at least 5 different learners speak during DQB closure; (3) answers reference mathematics (for example, 'w appears as w times w, so the equation has w^2'); (4) learners distinguish linear (perimeter $4w + 4$) from quadratic (area). Note questions that stay unanswered and record them as follow-up.
Model Building Phase  VIDEO: Using the quadratic formula: number of solutions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Quadratics by taking square roots: with steps <i>Related topic: not an exact match for this lesson.</i>	Each group builds a final cumulative model as a poster (manila paper) or digital slide that maps the journey: (1) recognising quadratic expressions and the general form $ax^2 + bx + c = 0$ (hall: $a = 1$, $b = 2$, $c = -120$); (2) expanding with the distributive law/FOIL; (3) factorising; (4) solving by factorisation using the Zero Product Property; (5) completing the square; (6) quadratic formula and discriminant ($D > 0$ two solutions, $D = 0$ one repeated solution, $D < 0$ none); (7) checking in context and rejecting invalid solutions. One panel shows the Lesson 1 model (trial-and-error table, guess for w) beside the final model (equation, three methods, $w = 10$ m, length 12 m, 120 m²). Another panel shows a decision guide for choosing a method and	Say: 'Show how your thinking changed from the first lesson to now. Put your Lesson 1 model next to your final model.' Provide a checklist of the required panels. Circulate and ask: 'Where on your poster does the context choose the answer?' and 'Which method did you use for the verandah and why?' Give WAIT TIME of 10-15 seconds before prompting. Cold-call 4 groups to present their comparison in 1 minute each. Ask presenters to name one change in understanding, for example from 'guess and check' to 'form an equation'. During the gallery walk, ask visitors to confirm the hall answer: 'width 10 m, length 12 m'. Close with: 'Which problem in Githurai, Kitale or your own village could you now solve?' Give the exit ticket: solve $x^2 + 4x - 21 = 0$	Model Comparison (Lesson 1 vs Final) and Gallery Walk: placing the first and final models side by side makes learning visible, while peer feedback in the gallery walk strengthens connections among expression, equation, method and context.	Observable indicators: (1) the Lesson 1 vs final comparison names at least two specific changes (for example, trial and error to equation; one guess to exact solution); (2) the poster includes all three methods with the hall answers $w = 10$ or $w = -12$ and the final answer of width 10 m, length 12 m; (3) the discriminant rule is shown correctly; (4) the context-rejection step appears explicitly; (5) the exit ticket $x^2 + 4x - 21 = 0$ gives $(x + 7)(x - 3) = 0$, $x = 3$ or -7, with -7 rejected for a width. Use a 3-level rubric (secure, developing, beginning) and plan follow-up groups for learners at beginning level.

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	one application (Kitale, tickets or verandah). Groups then do a gallery walk, leave one 'glow' and one 'grow' comment on sticky notes, and finish with a personal reflection: 'How can quadratic thinking help my community plan?'	by any method, justify the method, and state which solution would be rejected if x were a width.		
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D. TEACHER REFLECTION

Reflect after the lesson: (1) Which method did learners choose most often, and did their justification show real understanding of efficiency, or did they simply copy the quickest-looking route? (2) Could learners explain why a quadratic equation can have two solutions and why context rejects one ($w = -12$, $w = -6.5$, $x = -12$)? (3) Did learners clearly distinguish the linear perimeter $4w + 4$ from the quadratic area $w(w + 2) = 120$? (4) Which DQB cards were hardest to answer, and what does that suggest about gaps in the unit? (5) Did every learner contribute to the group poster? Note learners who need more practice on factorising when a is not 1 (such as $2w^2 + w - 78$) or on expanding $(12 + 2x)(10 + 2x)$, and plan short remediation. Record which Kenyan contexts (Githurai, Kitale, ticket fundraising) motivated learners most.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the Githurai hall equation $w(w + 2) = 120$, or $w^2 + 2w - 120 = 0$, gives the same solutions, $w = 10$ or $w = -12$, by factorisation, completing the square and the quadratic formula ($D = 484$). New problems gave Kitale land $w = 6$ m (length 13 m), Harambee tickets $p = 10$ or 20 KES, and a verandah of $x = 1$ m.
What did I learn?	We learned that a quadratic equation has the unknown multiplied by itself (highest power 2), unlike a linear equation such as the perimeter $4w + 4$, and can have two solutions. We learned how to choose the most efficient method and to check each solution in the context.
How does this explain the phenomenon?	We explain that the hall must be width 10 m and length 12 m (area 120 m ²) because $w = -12$ is impossible for a width, so context decides which solution is used. Quadratic thinking lets the Harambee committee and our community find unknown dimensions and quantities for real planning problems.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.